

Mathematical Neuron-Neural-Networks Dynamics Modeling and Brain Complexity of Multicriteria Fractal-Based Ensemble Predictive Algorithmic Systems

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ABSTRACT

Mathematical dynamical ensemble modeling of neuron and neural network dynamics entails the precise capturing of the functions and complexities in multi-criteria decision-making processes while being capable of inferring and reconstructing the dynamics and structure of complex entities by serving the observability and evaluation emerging amid different flows of information. The complex dynamic system of the human brain encompasses the connections across neurons at various levels of granularity and connection characteristics. Stroke, one of the causes of morbidity and mortality worldwide, is among the most debilitating neurological diseases having a possible risk of loss of motor or cognitive function, causing long-term disability. Thus, stroke, which displays profound complexity and dynamic structures, requires timely intervention and heightened sensitivity in terms of capturing subtle details and findings. For predictive purposes in health-care systems under conditions dependent on time and highly complex, computing systems and Artificial Intelligence (AI) applications, with resilient, complex and informative properties, enable realization of intelligent properties alongside with facilitation and empowerment for accuracy in identification, detection, diagnosis, plan of treatment, follow-up processes and outcome prediction, where human knowledge, reasoning, intuition, experience and heuristics are essential. On the other hand, multifractal methods are utilized efficiently for the analyses of various complex systems with heterogeneous agents and investigation of multivariate datasets. In healthcare, it is essential to quantify the nonlinear brain dynamics and its anatomical structure to grasp its subtle features, and yet, since brain operates on multiple time-scales, this poses a formidable challenge considering its characterization across different interacting spatio-temporal scales involved. Characterized by a high degree of diagnostic complexity, neurological diseases require accurate prediction, individualized treatment and timely management. Across these strands, current study aims to facilitate multifaceted decision-making, optimized outcome prediction and tackle the complexity of stroke, which can result in accurate prediction and efficient management of the disease. To this end, the mathematical model proposed in this study has adopted the subsequent stages: (i) multifractal regularization method is applied to the stroke dataset (X), and by this application, the mf-stroke dataset ($\hat{X}$) has been generated. (ii) ten different algorithms composed of deep learning methods, ensemble methods, fundamental methods and clustering methods are applied to the stroke dataset and generated mf-stroke dataset ($\hat{X}$) (iii) the comparison of predictive accuracy rates is made (along with the specificity, sensitivity, precision, F1-score and Matthews Correlation Coefficient analyses). Considering the gap in the literature where these methods we proposed have been used in an integrated way by multifractal analysis, the results of the study

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have demonstrated that the stacking method has yielded significantly more accurate results, and singularity analysis, as a multifractal method, has been performed on the mf-stroke dataset ($\hat{X}$). The results of the mathematical model proposed in this study validate the importance of deep learning methods, ensemble methods (bagging methods, boosting methods and stacking method) and fundamental methods as well as clustering methods in AI along with the singularity analysis by multifractal regularization. Consequently, it is projected that through the proposed model with its approach-wise novel aspects, predictions and treatment plans can be performed efficiently owing to the mapping capabilities and learning ability achieved by the integration of AI methods. Predicting the stroke subtype accurately is vital in healthcare for the patient outcomes including life-quality, (self-) control and survival. Given these aspects, the established mathematical model proposed herein is projected to be beneficial and advantageous for the accurate classification, evaluation, multicriteria decision-making and prediction to address uncertainty, variability and unpredictability.

1. Introduction

Fractal mathematics, whose research object is the complex natural phenomena and processes, is employed for investigating, analyzing and modeling of complex and irregular patterns as well as structures, which is essential in capturing the infinite complexity of nature. Amid such scope is the role of fractal dimension as a significant parameter to be resorted to for estimating the roughness of surfaces and / or other patterns, with fractal dimension of man-made targets being lower than that of the natural objects, which suggests its feasibility in differentiating different targets and objects by fractal dimension calculation in instances including mode recognition, decision-making, image processing and feature classification [1]. Succinctly put, fractal mathematics is employed to model complex and irregular structures with fewer data points, while fractals refer to the common term in mathematics, which manifests self-similar complex geometric shapes with patterns that have fragmented dimension. With the main features, of self-similarity, irregularity and iterative nature, the use of fractals comes to the fore to measure specific phenomena which display challenges for the description of objects which are self-similar or which tend to repeat themselves on different scales. Moreover, the self-repeating and infinitely detailed patterns of fractals at ever-diminishing scales transcend the realm of pure mathematics, which renders the gap between abstraction and reality paving the way from the natural world toward advanced technological breakthroughs. These features show the quintessential example of how simplicity generates complexity in addition to how it is possible that patterns can emerge from randomness. Bearing these in mind, it can be noted that fractals are not merely a theoretical construct, but they also constitute a passage through which the world can be seen and constructed, which can be exemplified by the metaphor of a bridge between the abstract and the concrete as well as the known and the unknown.

Mathematics and fractals manifest themselves through three mathematical properties which can be stated as self-similarity, fractional dimensionality and recursive generation, and owing to these, numerous applications of fractal geometry are possible across many fields (i.e. medical imaging and biomedical analysis, artificial intelligence (AI) and pattern recognition, feature detection, scientific computing and engineering, and so forth) [2]. As the broad application areas indicate, fractals cannot be only regarded as mathematical constructs, but they also display their function by mirroring real-world complexity when the ability of nature in terms of self-organizing, adapting and growing in self-similar patterns is under consideration, which used to be the prototype of the AI models' inspiration in the early periods of their development. As the human brain possesses fractal-like structures, the way neurons branch out is akin to fractals, and in that regard, the deep neural networks of AI originated based on this complex and efficient pattern of branching. These concepts are carried to a further level by the use of recursive structures for the simplification of network architecture without comprising the performance element. Beside these benefits regarding neural networks and biological fractals, efficiency in natural systems is also observed since fractals optimize resources in nature, while AI applications translate such efficiency in the processing of complex tasks through mimicking fractal structures so that more reliable, efficient and scalable algorithms can be set.

Multifractals, as an extension of fractal geometry, are designed to describe systems that possess multiple scaling behaviors instead of a single fractal dimension, and thus, multifractals can capture the complexity and variability of scaling properties across different regions of a system, whereas uniform self-similarity characterize traditional fractals. The capability of multifractals capturing heterogeneity within scaling properties is through measures such as Hölder exponent to quantify variability and multifractal spectrum [3]. From the viewpoint of systems thinking, a multifractal system refers to a generalization of a fractal system where a single exponent, or the fractal dimension, falls inadequate for describing its dynamics. Rather, a continuous spectrum of exponents, namely the so-called singularity spectrum, is to be required [4]. Consequently, multifractal analysis is employed for investigating datasets, frequently along with other methods of fractal as well as lacunarity analysis, and the technique requires the distortion of datasets extracted from patterns so that multifractal spectra can be generated, illustrating how scaling varies over the dataset. Hence, multifractal analysis has been used to interpret the generating rules and functionalities of complex networks [5] along with a broad array of practical incidents like earthquakes' prediction and medical images' interpretation [6–8].

Types of biological neurons comprise sensory neurons that are activated by sensory physical or chemical input from the environment, motor neurons which transmit impulses from the spinal cord to skeletal and smooth muscles as well as interneurons connecting spinal motor and sensory neurons besides communicating with one another, which enables forming circuits of various complexity. Biological neuron models also referred to as spiking neuron models signify the mathematical descriptions of electrical signals' conduction in neurons which are electrically excitable cells in the nervous system capable of firing electric signals across

a neural network, which is known as action potentials or spikes. Mathematical models as such describe the neurons' biophysical and geometrical characteristics concerning their impact on the conduction of electrical activities [9]. Spiking neurons happen to be significant processing unit of the nervous system, and one example of a spiking neuron model could be a detailed mathematical model encompassing spatial morphology, whereas another one could be a conductance-based neuron model viewing neurons as points and describing the membrane voltage dynamics as a function of trans-membrane currents. "Integrate-and-fire" model is known to significantly simplify the description of ion channel and membrane potential dynamics [10]. Furthermore, Josephson junctions by nature are neuromorphic hardware devices that can mimic the excitability and spiking dynamics. When combined with other superconducting elements, they are capable of emulating additional behaviors that are existent in biological neurons [11]. Josephson junction neurons reproduce characteristic behaviors of biological neurons like action potentials, firing thresholds and refractory periods. Relatedly, superconducting circuits that involve Josephson junctions that model biologically realistic neurons are presented with the demonstration of a superconducting circuit model of a biologically realistic neuron with significant speed and efficiency concerning neuromorphic computing. The work pointed toward a new approach and tool regarding the simulation of large-scale neural networks and the exploration of their long-term dynamics [12]. It is essential to understand the neuron behavior to advance power-efficient neuromorphic computing, and concerning this end, another work represents a novel model of the spiking neuron obtained through the combination of a threshold-switching selector with a tunnel diode [13]. The study's novel bio-inspired neuron model proposed uses a threshold selector and a tunnel diode, which specifically focuses on the ability to modulate its excitability. It investigates the circuitry, identification of models concerning the basic nonlinear elements, which demonstrates the capability of the neuron for excitability modulation, suggesting that the neuron's firing threshold and behavior can be dynamically adjusted, as a noteworthy characteristic of biological neurons, essential for complex neuromorphic systems. One other related work explores the new bio-inspired modes of operation of a conventional $2JJ$ superconducting model of a biological neuron, indicating the future direction for extending the applicability of superconducting hardware models in terms of mimicking other types of synaptic plasticity like spike-timing-dependent and long-term plasticity, which may pave the way for energy-efficient and fast hardware embodiment of artificial intelligence [14]. Another biological neuron model is the Hindmarsh-Rose model which is reported to provide a high level between variability and complexity. One relevant study focuses on the model parameter identification problem which is a significant part of control system theory and explores the Hindmarsh-Rose model parameters identification problem with measurable output [15]. This problem is addressed by the developed algorithm based on speed gradient method, while the network identification problem is handled by the proposing of an identification algorithm able to estimate the model parameters and coupling strength.

Besides these, map-based neuron models have been acknowledged to be valid phenomenological neuron models owing to their computational efficiency and flexibility to generate rich patterns [16]. Correspondingly, a new chaotic path-planning algorithm based on the Courbage-Nekorkin (CN) neuron model is developed in a work for reducing the number of iterations required for a mobile robot to uniformly cover a designated area [17]. The scheme proposed confirms its usefulness as a chaotic search or optimization engine, which is a pre-requirement for applications concerning chaotic metaheuristics for clustering. Regarding the implementation accuracy of a neuron described by simplified Hodgkin-Huxley equations in fixed-point arithmetic is investigated in the research in which the principle of constructing a fixed-point neuron model is described with different numerical methods [18]. The authors make use of the interspike diagrams and refractory period analysis for the experimental study of synthesized discrete maps of the simplified Hodgkin-Huxley neuron model through which they demonstrate that explicit midpoint method is better suited to simulate the neuron dynamics compared to the explicit Euler method that is in common use. Through these models, artificial neurons' capability to replicate the dynamic characteristics exhibited by their biological counterparts have been revealed and demonstrated [19].

The human nervous system is made up of a large quantity of linked neurons referred to as neural network. Neurons and glial cells are the two primary types of brain cells, while the spatial distribution of the brain cells comes with two main types of brain tissue matter, which are the white and gray matter [20]. Neurons, as the electrically active nerve cells, maintain an electrical potential across their membranes by means of ion pumps. Axon, soma, dendritic tree and synapses are the four functionally relevant components [21]. Nervous and network make up the terms to describe a network's structure similar to a graph, which depicts the artificial neural network (ANN) as a kind of computer system inspired by biological neural networks. While the nodes in a neural network conduct basic calculation, the connections carry signals across the nodes [22]. Neurocomputing signifies the utilization of computer simulation technologies geared toward exploring complex mathematical neural models, which locates it at the intersection of computational neuroscience and ANNs or deep learning. One of the primary features of neurocomputing is its capability to tackle problems that are too complex for rule-based algorithms. High-dimensional data with many variables as is the case in natural language processing and image recognition, unstructured data which do not fit appropriately into a dataset table or nonlinear relationships where relationship between variables is not in a straight pattern may be among the related problems in question. Given these aspects and challenges, neurocomputing handles these tasks through finding patterns and relationships in complex data. With these considerations, brain modeling is elevated to a quantitative physical theory by combining data, theory, experiments as well as computation. The advent of magnetic resonance imaging (MRI) has brought about advancements in medical practice by incorporating a reasonably-high resolution and non-invasive means of observing the brain's internal states. Thus, MRI imaging has become an essential source of information for clinical neuroscience as well as research into neurodegenerative diseases due to its ability of mapping and measuring properties of the cortex and white matter, along with determining the patterns of water flow within the brain and isolating regions of high cognitive processes.

The other remarkable advancement has been observed with the rise of computing power, which has allowed for modeling and simulating the responses of three-dimensional structures through finite-element modeling. Although the original methods were utilized to assess the safety of building structures, they were adapted to biology by simulating the responses of the heart, arteries, bones, lungs, and eventually, the brain which is a complex material with captivating properties not shared by any other organs [21]. Since

the generation of physiological finite element meshes of the brain is challenging, and the sulcal and gyral folds of the cortex are intricate, mathematical modeling of the human brain and its anatomical mesh generation come to the fore. As a consequence, the ability to accurately capture anatomical features could ensure the addressing of many human health-related problems, particularly when understanding the mechanisms underlying neurodegenerative pathology evolution is at stake.

Apart from mathematical modeling, neural network modeling usually has to do with the stimulus-driven responses, and most of the activities going on in the brain are internally generated. The exploration of such aspects in neural network dynamics is vital to understanding the way neural circuits generate cognitive function [23]. Correspondingly, neural-neuron network dynamics studies how neurons, as the basic units of the nervous system, interact with one another in the networks to process as well as to transmit information. To these aims, neural-neuron network dynamics dwells on investigating the neurons' chemical and electrical activities in conjunction with their synapses (connections) and the neural networks' emergent behaviors. The preliminary principles into such exploration include neuron dynamics which examines individual neuron behaviors and network dynamics that quests how groups of neurons interact, synchronize and form patterns beside different phenomena such as oscillations and plasticity. These uses derived from neural dynamics lend themselves to applications in various fields, some of which are medicine, neuroscience and AI.

The connection between neural dynamics and complexity makes itself evident through the examination of the way intricate patterns and behaviors emerge as a result of the interactions of neurons within the networks. Due to these factors, complexity in neural systems frequently result from the interconnection of feedback loops, nonlinear dynamics and the extensive connectivity of neurons. Given these, the study of complex systems aims at enhancing the capacity of comprehending universality that emerges when systems are highly complex with their interconnected constituents. Such understanding cannot be complete by only understanding the behaviors of the parts acting together to form the behavior of the whole. Grasping the interwoven elements and self-organization as the important qualities of complex systems should be at stake for a thorough understanding since complexity characterizes many evolving parts that interact with one another in varied forms, displaying nonlinear patterns as aggregate [24].

Neural processes driving different functions such as perception, attention, coordinated movement, intelligent acts and reasoning result from the processes related to learning and memory, all of which are not necessarily conscious and explicit, since what happens in the brain implicitly supports rather than explicitly reflecting the conscious processes of thinking. Nevertheless, it is presumed that all these relevant processes must arise from the calculations carried out directly by neurons or related synchronized population ensemble effects displayed through the field surrounding the neurons [25]. As a complex network whose ensemble time evolution is directed by cumulative interactions of its cellular components like neurons and glia cells, the human brain is highly complex and intricate. Coupled through chemical neurotransmission and receptor activations, the distinct elements' interaction with one another happens at varying degrees, which is triggered by different cellular activities from internal biological reconfigurations to external interactions with the agents of other networks [26]. Considering these interactive complexity and heterogeneity, ensemble methods are acknowledged to be among the powerful means of machine learning combining multiple models for enhancing predictive performance as well as robustness. Rather than relying upon a single model, ensemble methods aggregate the outputs of several models, which are often called base learners or weak learners so that it can be possible to establish a more accurate and reliable final model. Consequently, ensemble algorithms have emerged as robust computational techniques to interpret multifractal behavior within the given datasets. Through the aggregation of predictions or solutions from diverse models, ensemble methods are employed toward the aim of improving accuracy. The ensemble techniques and ensemble neural networks, therefore, align with the multifractal systems' stochastic nature, which allows for adaptive learning across multivariate patterns. The capability of multifractal analysis lies in detecting spatial-temporal variability and the synthesizing feature of ensemble methods enhances predictions from diverse models. In a nutshell, fractal and multifractal frameworks provide the required theoretical foundations, whereas ensemble algorithms are used for data analysis through identification, detection, classification and prediction of emergent behaviors.

Mathematical modeling of neurons and neural networks within complex network dynamics as well as computational principles including the elements of fractal generation have been extensively explored in research. In addition to these, the consideration of the intricate and inherent heterogeneity - dynamical morphology in complex systems, fractal and multifractal analyses are conducted to efficiently measure different phenomena in diverse fields like economy [27–29], signal processing (EEG/ECG) [30–32], diagnostic imaging [33,34], epidemiology [35–37], space science [38], geology [39–41], materials technology [42] and brain structure [43], neurology / clinical neurosciences [44,45] and other related fields [46], to name just a few. Fractals are ubiquitously present in the universe, ranging from natural objects to universal structures, in the most minor item as well as at the largest scales. Brain, as a complex organ with billions of neurons and components in micronetworks and macronetworks [45], is endowed with full interactivity and interplay, which could make it be handled as a fractal object. Within this framework, fractal and multifractal-based quantification becomes essential as the brain operates on multiple scales from microscales and macroscales. Fractal geometry, on the other hand, serves for the quantitative analysis and description of the natural systems' geometric complexity in this domain which encompasses unknown, that is to say, hidden variables. At this point, fractal and multifractal methods provide robustness in complex systems. To illustrate, Artificial Intelligence (AI) with expert system applications is an emerging field of computer science which aims at imitating and emulating cognitive processes and human intelligence in machines via certain techniques based on mathematics. The importance of this aspect is observed in the diagnosis and management of various diseases, by revealing the varied opportunities and future potentials for care, minimization of risks and maintenance of social life while uncovering the hidden and underlying threats and risks. AI and related fields such as neuroscience, brain related studies in complex dynamical systems bring together different aspects of the domains and drive each other, yielding benefits for both.

There are different diseases related to the structure, function and morphology of the brain which include malformations, psychiatric disorders, pathologies, demyelinating diseases, brain tumors and neurodegeneration. Stroke is one of the leading reasons for morbidity and mortality worldwide, being one of the most debilitating neurological diseases with a possible risk of loss of motor or cognitive

function [47]. Having different subtypes makes stroke more complex to be diagnosed and managed considering the medical condition. Ischemic stroke is the most common type of stroke in which an inaccurate intervention may lead to critical consequences for the patient's functionality and life. Under these critical circumstances, modern neurosciences based on a holistic perspective, necessitate accurate, robust, applicable and reliable solutions to minimize risks and maximize health benefits (for further details on stroke subtypes, please see [44,48,49]). For instance, magnetic resonance imaging (MRI) has emerged as a powerful means for the diagnosis of stroke, providing detailed images of the brain structures as well as abnormalities. Yet, it is possible that the analysis of MRI data pose challenges as a result of its complexity, which makes the need for efficient and reliable classification algorithms essential, specifically when one aspires to understand the dynamics of the brain [50]. While the previous studies focused on the classification of stroke by means of medical images [51,52], the approaches in more recent studies have concentrated on the extent of lesions and limited correlation to functional damages [53]. More recently, the inner functioning of the brain based on the impact of one brain on another one as well as the way lesions compromise the related interactions [52,54]. Complexity related to the network systems and the brain's connectivity encompass the complicated interactions across neurons and network thereof. This sort of connectivity, within this domain, is categorized into structural, functional and effective connectivity, which helps with the profound exploration of the brain's inherent workings [55]. These features intermittently require the use of sophisticated whole-brain modeling for the prediction with regard to the best ways to probe the transitions between different brain states. Accordingly, computational neuroscience has encountered the querying of new ways to strike a rebalance between the complex dynamic system of the human brain, which have resulted in robust frameworks for determining the functional hierarchical organization that can describe any given brain state within the dynamics of interacting neuronal and neurotransmitter systems [56].

Stemming from the aforementioned developments, advancements in AI with computing system applications provide the path for timely and efficient decision-making, big data analysis and so forth, which play a critical role in medicine, particularly for individualized treatment courses. Deep learning (DL), by the same token, has the capability of learning the high-order and multi-feature abstract data. In this framework, Artificial Neural Networks (ANNs) act as a subtype of deep learning, emulating the biological neurons with input and minimum one hidden layer as well as an output [57]. AI serves the purpose concerning accurate, robust, applicable and reliable solutions in complex dynamical networks by achieving more accurate results based on big data-driven research. DL methods utilize a specific type of AI architecture, called ANNs, emulating the brain functions of humans. Thus, ANNs are acknowledged to be at the core of several of the greatest advances in modern technology, enabling significant progresses in applications ranging from computer vision to speech recognition as well as autonomous processes, and many more [58]. Each layer consisting of neurons connected to others in the subsequent layers, ANNs are made up of an input layer, multiple hidden layers and one output layer. By employing a set of quantitative variables as the input to the first layer, the corresponding layers used together are for learning and optimizing related to the neurons in the network [59,60]. ANNs are also useful for solving the classification problems based on this attribute so for enhanced image quality, diagnostic accuracy and decision making in treatment can be realized through the cooperation of AI and DL methods [59].

It is acknowledged that no single algorithm can prove to be superior in all potential cases, which led to the idea of ensemble learning. As the statistical and computational learning procedures, ensemble methods suggest human social learning behavior related to looking for multiple opinions prior to a critical decision [61]. Ensemble member makes the classification of the input data and afterwards individual predictions are combined by a specific algorithm, which is named a combiner, so that the prediction of the entire ensemble can be formed [61]. The upper hand of ensembles is that a group or an ensemble of classifiers can learn a more complex decision and hypothesis with higher accuracy by splitting different classes compared to one classifier individually. During this multi-layered process, there is a trade-off in question between diversity and reliability, which suggests a compromise between diversity of individual predictions and the accuracy thereof [61–63].

Performance accuracy is one of the most formidable challenges in scientific inquiry, and regarding computational concerns, AI is one of the agents which enables one to achieve more realistic outcomes. In critical decision making under uncertain circumstances, accuracy becomes even more important. AI applications can be utilized to provide actionable insights and optimized outcome prediction for the acceleration of the diagnosis and ensuring of an accurate medication and intervention procedures timely following the onset of the stroke. Considering the complex structured and critical neurological disorders and other brain-related medical problems, timely, accurate and adequate medical care is to be delivered so that patients can regain their mobility and skills thoroughly, with more gains being achieved during the follow-up and post-treatment as well as rehabilitation processes while reducing mortality and / or morbidity. Specifically, for stroke cases, AI can provide assistance regarding the different aspects of stroke treatment regime including but not limited hemorrhage, infarct, detection, large vessel occlusion detection, classification and segmentation [64]. There are many studies in which AI plays a role to provide solutions in uncertainty in different subfields of medicine. The study of [65] provides a review of artificial intelligence focusing on its applications in radiology, pathology, ophthalmology, and dermatology, presenting the possible uses and applicability aspects of AI in a setting equipped with technological advancements. Regarding the requirement of accuracy, the study by [66] offers a description of an information system named Sechenov Datamed which solves the concern of supporting medical decisions depending on evidence-based medicine through artificial intelligence methods. Their work helps reduce the medical errors in the field. Advances in the applications of AI from experimental stages to the implementation phase are encouraged by increasing data in reproductive medicine as well (please see [67] for further studies in the related area [68–71]). The study by [72] is concerned with the incorporation of AI and machine learning models for infertility. Individualized medicine and precision are two important elements so that the unique characteristics of the individual patient are taken into consideration for evaluation and diagnosis. One review study concerning expert systems applied in different medical domains provides the heterogenous merit of the solutions proposed in the literature so that possible benefits can be obtained from the systems applications. It is also stated that some criteria need to be followed by medical expert systems, ranging from fuzzy logic to wearable solutions [70]. Correspondingly,

another study presents a novel expert system reliant on infrared thermography and thermal analysis pertaining to facial skin and fingertips, along with a rule-based method as well as heuristic knowledge for the classification and diagnosis of human stress students [71]. Stroke is one of the diseases which requires this kind of individualized aspect and sensitivity. Regarding the studies in which AI and stroke are handled, the study by [73] deals with the prediction of symptomatic intracerebral hemorrhage after the thrombolysis in acute ischemic stroke. Their prediction is based on laboratory, imaging and clinical data through the multiple ANN models which are multilayer perceptron (MLP), radial basis function (RBF), support vector machine (SVM) and probabilistic neural network (PNN). By taking sensitivity into consideration, the study suggests the generalizability of the ANN analyses.

Regarding stroke detection, lesion segmentation and precise localization of abnormal elements in stroke disease, deep learning algorithms have had a significant impact as developed deep learning models save time and enable more accurate results [74]. Accordingly, the study by [59] used deep learning for acute stroke patients in the image-level detection of arterial occlusions in four-dimensional CTA (4D-CTA) imaging. The deep learning approach in the study with the presented method achieved high levels of sensitivity and specificity, at 95% and 92%, respectively. In another study conducted by [75], four machine learning algorithms, which were Bootstrap decision forest, Boosted trees, Logistic Regression and Deep neural network, were employed for the prediction of early neurological deterioration in acute minor ischemic stroke. The results of the study suggest that the machine learning algorithms integrating clinical and neuroimaging information can be used for efficient prediction and accurate design of treatment strategies to achieve more functional outcomes. The study performed by [76] proposes a new framework relying on the synthesized pseudo Diffusion-Weighted Imaging (DWI) in order to get better image quality for accurate segmentation. The proposed framework of the study is found out to have potential to enhance the diagnosis and thus the treatment related to ischemic stroke. With regard to acute and subacute stroke lesion segmentation, a deep learning methodology was proposed in the study by [77]. More improved segmentation outcomes were achieved through the use of a combined approach so that the impacts of class imbalance would be minimized by obtaining a less biased segmentation and better balance concerning specificity and sensitivity. In another study by [78], a stacked ensemble model with XGBoost metalearner achieves was developed for the classification of motion artifacts in brain scans, which yielded robust prediction of rigid-body motion severity. The clinical significance of the motion artifacts lies in the sensitivity of stroke towards them. Another research work [79] addressed ensemble predictive method with L1 regularization, random forests, multivariate adaptive splines and gradient-boosted decision tree ensemble (XGBoost) were used. The findings emphasize the phenotypic heterogeneity of embolic strokes of undetermined source (ESUS) along with the personalized approaches in stroke treatment. The economic benefits of brain stroke detection by data-driven approaches were emphasized in the subsequent study [80] which indicated that the use of ensemble of multiple brain stroke prediction models could be helpful in improving the prediction performance. The experimental results of that study revealed that the ensemble of the prediction models with majority voting approach could yield better performance outcomes compared to the individual prediction models.

Furthermore, high degrees of complexity characterize neurological diseases in which accurate prediction, individualized treatment and timely management of the course are all of critical importance. Accordingly, stroke addressed in this study is concerned with the brain which displays profound complexity, heterogeneity and dynamic structures across temporal and spatial scales. The integrated multistage mathematical model proposed by this study aims to enable the intersection, in AI, along with the singularity analysis to extract subtle patterns in the stroke dataset (X) by multifractal regularization (L2 normalization). When the earlier studies in the literature are reviewed, there appears to be a gap regarding the research in which an integrated multistage mathematical model with algorithms, made up of deep learning methods, ensemble methods (bagging methods, boosting methods and stacking method), fundamental methods and clustering methods as well as multifractal regularization (L2 normalization) have been employed on multifeatured complex stroke dataset (X). Considering the stroke dataset (X) addressed and the complementary methodology proposed, this study has the following steps with the relevant contributions. To start with, multifractal regularization (L2 normalization) method has been employed to the stroke dataset (X) (see Table 2) and through this application, the mf-stroke dataset ($\hat{X}$) was generated. Thus, the distance was minimized and calculations were simplified with respect to the dataset with a high degree of complexity, which is one significant contribution. Secondly, the algorithms, comprised of deep learning methods, ensemble methods, fundamental methods and clustering methods were applied to the stroke dataset and the mf-stroke dataset ($\hat{X}$). The related contribution of this stage in the proposed method is to assist the accurate individualized treatment and intervention for stroke cases characterized by complexity and unpredictability by enabling the timely and accurate determination of the stroke subtype. As the final stage, the accuracy rates have been compared (considering the specificity, sensitivity, precision, F1-score and Matthews Correlation Coefficient analyses) following the deep learning methods, ensemble methods, fundamental methods and clustering methods (see Table 7) indicated in the second stage. As a result of these interconnected procedures, it has been demonstrated that stacking method, among the related ensemble methods, has yielded more accurate results since the singularity analysis has been performed on mf-stroke dataset ($\hat{X}$), through which significant and self-similar attributes have been able to be detected. The results of our proposed method in this study demonstrate the importance of the intersection of deep learning methods, ensemble methods, fundamental methods and clustering methods in AI along with the singularity analysis by multifractal regularization. Thus, rigorous advancements in artificial intelligence with advanced computing system applications aside from mathematical model provide the path for efficient and timely decision-making, big data analysis, optimized outcome prediction and so on, owing to properties which require human knowledge, intuition, experience and heuristics, while at the same time play a significant role regarding the individualized treatment courses for neurological diseases like stroke. Another significance of the proposed modeling paradigm of our study is that the multistage mathematical model has the capability of achieving accurate results on the different tasks, such as classification, clustering and prediction, concerning the multi-feature dataset with hidden data characterized by high degrees of complexity, which is stroke in our case.

1.1. Motivations of the integrated multistage mathematical model proposed

Degrees of complexity in different areas, including modern neuroscience have required the emergence of applications to be more intelligent in the present period. Accordingly, complex systems research and artificial intelligence (AI) with computing system applications exhibit a close link with each other, providing a common denominator for an extensive array of interdisciplinary interactions. Within this framework, the integrated multistage proposed mathematical model proposed in the current study aims to enable the intersection of deep learning methods (Feed Forward Neural Network, Multi-layer Perceptron), ensemble methods (bagging methods (Random Forest and Extra Tree classifiers), boosting methods (XGBoost, Gradient Boosted, Light Gradient Boosting classifiers) and stacking method) and fundamental methods (Decision Tree, Logistic regression) and clustering methods (K-nearest neighbor, Support Vector Machine) in Artificial Intelligence (AI) along with the singularity analysis to extract the subtle patterns in the stroke dataset by multifractal regularization. As is the case with other diseases, complexity, heterogeneity and uncertainty components are common in the cases of neurological diseases like stroke, which are characterized by high degrees of complexity, dynamic and varying structures, courses, processes of follow-up, treatments as well as patterns. Accurate, optimal, secure and safe algorithms provide the facilitation of medical processes, which in turn facilitate the physicians and maintain the life quality of the patients through accurate information relay and cost-effective means. Moreover, resilient advancements in AI with the integration of appropriate mathematical models can enable efficient and timely decision-making and big data analysis.

One of the novelties aimed with this paper is related to the observation that there is a gap in research where a multistage integrative mathematical combination approach with AI, algorithms, made up of deep learning methods, ensemble methods (bagging methods, boosting methods and stacking method), fundamental methods and clustering methods as well as multifractal regularization method have been employed on the multifeatured complex stroke dataset. Multifractal regularization (L2 normalization) method has been employed to the stroke dataset in this study which furthermore reveals that multifractals enable the distinguishing of the hidden details from the entirety of the object. Through the application of multifractal regularization (L2 normalization) method, the mf-stroke dataset has been generated. The advantage of this application has been to generate new results, while being able to minimize and simplify the calculations as per the stroke dataset that displays a high degree of complexity, which can be regarded as another novelty and contribution of the current study.

Our proposed mathematical model highlights the multistage and integrative deep learning methods, ensemble methods (bagging methods, boosting methods and stacking method), fundamental methods and clustering methods in AI besides the singularity analysis by the multifractal regularization method (L2 normalization). As a result of the applications (see [Table 7](#)), the mf-stroke dataset to which one of the ensemble methods, stacking, has been applied yielded the highest performance accuracy rates for the accurate classification of the stroke subtypes. Stacking method owns predictive power and generalization capability, and this study also has put forth the superiority of stacking method since singularity analysis has also been performed by the multifractal regularization method (L2 normalization), deep learning methods, ensemble methods, fundamental methods and clustering methods as the AI algorithms applied in our study can assist for the earlier detection of stroke and determination of the stroke subtype, which can also enable the acceleration of the treatment and improvement of the related outcomes depending on the treatment course and follow-up processes by facilitating accurate decision-making and optimized outcome prediction in terms of prescribing the right medication and appropriate intervention.

The stroke dataset addressed in the present study displays a multi-feature characteristic, for which heterogeneous methods have been opted for in our proposed method due to their presenting of different views regarding the data at hand. The advantage of this complementary and multistage methodology lies explicitly in capturing the dependencies between the classifier predictions. Our proposed mathematical model, with these properties, along with the algorithms applied in this category show that heterogeneous approach has yielded more accuracy, stability, effectiveness and efficiency. It may sometimes be observed that the subtle details related to stroke may be disregarded during the regular and conventional treatment procedures, which eventually leads to dire consequences for the patients. In the integrated multistage proposed mathematical model of our study, multifractal regularization method (L2 normalization) has been used to analyze singularities for the extraction of the subtle and hidden patterns in the stroke dataset, adopting two different approaches with the related stages. This can be considered to be another contribution of this study through a different and novel outlook of AI with expert and computing system applications when the critical significance of establishment and maintenance of individualized treatment and timely management of the stroke and its subtypes which manifest characteristics of vagueness, variability and uncertainty. All things considered, the use of multistage mathematical model in our study stems from the fact that accuracy is not the mere parameter to evaluate the significance of a particular technique, so multi-criteria decision making should also be considered, which constitutes the core of the complementary and integrative framework proposed.

The rest of the paper is organized in the subsequent way. [Section 2](#) deals with Materials and Methods, presenting the Patient Details and Complex Feature Engineering. Moreover, [Section 2](#) includes Methods, which include Multifractal for singularity analysis and Ensemble Predictive Methods along with Complex Feature Engineering. In [Section 3](#), Experimental Results and Discussion are described as in the integrated multistage approach proposed deep learning methods (Feed Forward Neural Network, Multilayer Perceptron), ensemble methods (bagging methods (Random Forest and Extra Tree classifiers), boosting methods (XGBoost, Gradient Boosted, Light Gradient Boosting classifiers) and stacking method) and fundamental methods (Decision Tree, Logistic regression) and clustering methods (K-nearest neighbor, Support Vector Machine) (see [Table 4](#)). In the last section, Conclusion and Future Outlooks are presented.

Table 1
Stroke patient details by age.

Stroke Subtypes / Age Ranges	0–29	30–36	37–49	50–70	71–90	91–104
Cardioembolic	18	5	21	195	391	59
Small Vessel	14	1	7	113	85	8
Cryptogenic	22	12	53	228	196	17

Table 2
The stroke subtypes handled and their details.

Stroke subtypes (number of patients)	Stroke subtypes (main heading of attributes)	Data size
Cardioembolic (689) Small vessel (228) Cryptogenic (528)	Medical history (HTN, hyperlip, DM, H/O stroke/TIA, AtrialFib, CAD, CHF, PAD/carotid disease, tobacco, ETOH)	1445 × 23
	Demographic information (Age, gender)	
	Treatment and medication data (Statin, antiHTN, antidiabetic, antiplatelet, anticoagulation, CT perfusion, neurointervention)	
	Results of laboratory test (mRS 90 days, hemorrhagic conversion, NIHSS admission, TPA)	
TOAST Main heading of attributes: type/etiology of stroke; HTN: hypertension; DM: diabetes mellitus; CAD: coronary artery disease; ETOH: alcohol; TIA: ischemic attack; AtrialFib: atrial fibrillation stroke; CAD: coronary artery disease; CHF: congestive heart failure; PAD/carotid disease: peripheral artery disease; NIHSS 90 days: National Institutes of Health Stroke Scale 90-day mortality; CT perfusion: computer tomography perfusion; NIHSS discharge: antiHTN: antihypertensive drugs after acute ischemic stroke; H/O stroke/TIA: history of transient ischemic attack; National Institutes of Health Stroke Scale		

2. Materials and methods

2.1. Patient details and multifeature stroke dataset

The stroke dataset investigated in this study is composed of 1445 individuals diagnosed with absolute clinic stroke subtypes at the University of Massachusetts Medical School and Stroke Services at UMass Memorial Medical Center (Worcester, USA). The related stroke subtypes include Small Vessel (228), Cardioembolic (689) and Cryptogenic (528). The ethical approving of the study has been done by the University of Massachusetts Medical School and Stroke Services at UMass Memorial Medical Center, USA.

The breakdown of the multifeature stroke dataset with respect to age details of the patients belonging to one of the three stroke subtypes is presented in Table 1 below.

The NIH Stroke Scale (NIHSS) is a widely utilized diagnostic method to evaluate the severity of the cognitive effects of a stroke patient's experiences. It is a quantitative measure about the neurologic deficit as a result of stroke, which is also used for the determination of appropriate treatment and prediction of patient outcomes. Table 2 presents the breakdown of stroke dataset attributes, i.e.: main headings of attributes, which are demographic information, medical history, results of laboratory tests, treatments and medication data (see further details in [44,49,81], belonging to three subtypes of stroke which are small vessel, cardioembolic and cryptogenic.

The baseline characteristics of the stroke patients in terms of attributes, statistics and descriptions as stratified by infarct side are provided in Table 3.

Having obtained the statistical aspect from the boxplot, this depiction reveals the relationship among the attributes for all the three stroke subtypes as per the complexity of the dataset. It is also observed that the distinctive features manifest more complexity due to the affinity of the boxplots that represent the stroke subtypes herein.

Table 3
Stroke dataset description.

Attributes	Status	Number of patients/values	Descriptions
Hemorrhagic con.	Yes	204 (0,09%)	Whether the ischemic stroke turned to hemorrhagic
DM	Yes	602 (27%)	Diabetes
H/O stroke/TIA	Yes	546 (25%)	History of stroke/TIA
AttrialFib	Yes	541 (25%)	Abnormal heart rhythm
CAD	Yes	513 (23%)	Coronary artery disease
CHF	Yes	229 (10%)	Congestive heart failure
PAD/carotid Disease	Yes	318 (14%)	Peripheral artery disease
HTN	Yes	1593 (72%)	Hypertension
Hyperlip	Yes	1197 (54%)	High levels of lipid (fat) in blood
Statin	Yes	1000 (45%)	Medications given to the patient are grouped into five broad categories
AntiHTN	Yes	1332 (60%)	
Antidiabetic	Yes	454 (20%)	
Antiplatelet	Yes	1031 (47%)	
Anticoagulation	Yes	242 (10%)	
CT perfusion	Yes	137 (0,06%)	Procedures used for treatment
Neurointervention	Yes	271 (12%)	
mRS 90 days	Low	2007	
	High	197	Dichotomized into low (0–2). high (3–6)
Tobacco	Yes	520 (23%)	Cigarette addict
ETOH	Yes	308 (1.7%)	Alcohol addict
TPA	Yes		Tpa (tissue plasminogen activator) is used to break down blood clots
NHSS admission		9,3 +/- 8,3	Measures the severity of stroke

2.2. Complex feature engineering

Complex systems include multiple interacting variables and emerging behaviors which are not possible to be predicted beforehand or comprehended merely from their related properties. Moreover, extracting the association between the components of complex systems is another challenge posed due to their non-linearity characteristics. Under such circumstances, if one deterministically focuses only one single aspect of a situation, it may not be possible to understand the whole system and get a unitary whole. Considering the emergent conditions, high levels of ambiguity, unstable and uncertainty in complex systems, certain engineering processes are required as a reality. At this stage, feature engineering and selection come to the foreground to for providing tools for the adjustment, reworking and re-representation of predictors so that the tools in the context can be placed within an accurate robust predictive modeling, which enables the uncovering of predictor-response relationships. Another stage of feature engineering involves the refining of a raw dataset. Through feature engineering, features for each label are built and the data which are used for the feature are filtered. Missing values, due to some human errors, privacy issues and interruptions in the data flow, seem to be one of the most frequently encountered problems for the preprocessing and preparation of the data for machine learning. Missing values can include information that might be incorrect, non-existent and/or incomplete, which are problematic since they reduce the machine learning algorithms' accuracy and performance. This problematic situation is known as the “curse of dimensionality” data, suggesting data with unrelated or redundant features [82]. To handle missing data, one viable solution is dimension reduction, and the other one is feature selection.

Most statistical and machine learning algorithms do not allow datasets with missing values and errors. As a result, it becomes necessary to handle the missing data. Some of the classical methods, like the univariate method, handle missing values to fill with mean, median, or mode. It might be a simple technique for missing value imputation, but it usually does not provide an accurate description of the data. To compensate this situation, complex feature engineering provides compensating complementary solutions in unstable environments with emergent structural and behavioral patterns, where emergence refers to shifts in the whole systems behavior. As has been addressed in this study, stroke is one of the diseases which manifests dynamic, not static, characteristics. For this reason, it cannot be known or predicted beforehand due to the emergence feature of this neurological disease.

In our study, k-Nearest Neighbor (kNN) algorithm has been applied to the stroke dataset (1445×23) in order to fill the missing values. In this way, lost data values have been handled more accurately compared to the univariate methods. k-Nearest neighboring (kNN) is an algorithm which classifies the neighboring data points through a measure of Euclidean distance. Thus, the missing values can be determined by using the completed values of nearby points. Fig. 1 illustrates the boxplots of 23 attributes shown against the output variable with their unscaled range on the y-axis and on the x-axis. For the enabling of computational efficiency and prevention of resource consumption, we have removed the correlated features generated so that it will not have a negatively impact on the model outcomes. To avoid peeking into the test set, we have computed the correlations based on the training set. (The use of automatic feature generation was avoided to form interactions, which causes “feature explosion” since all the possible

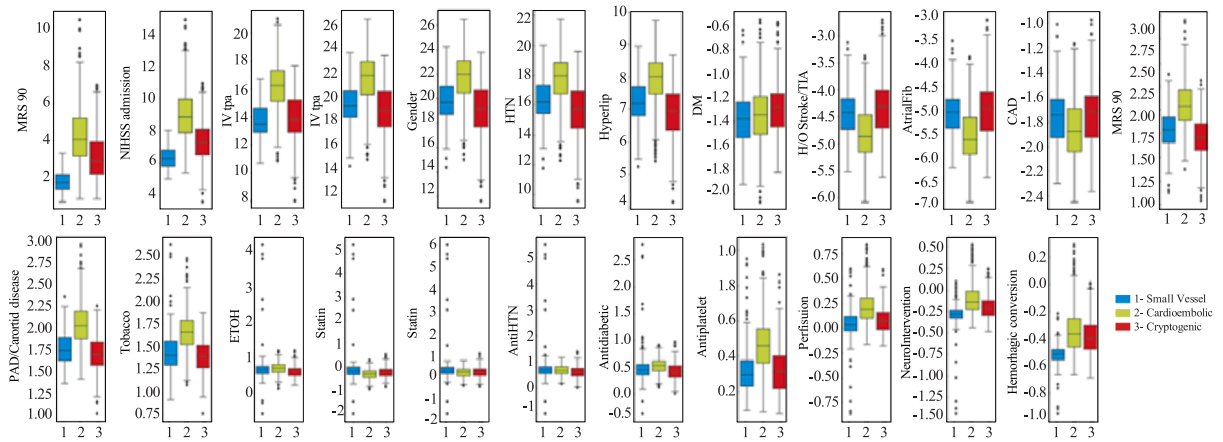


Fig. 1. Boxplots of 23 attributes are shown against the output variable with their unscaled range on the y-axis and on the x-axis 1, 2 and 3 represent small vessel, cardioembolic and cryptogenic stroke subtypes, respectively.

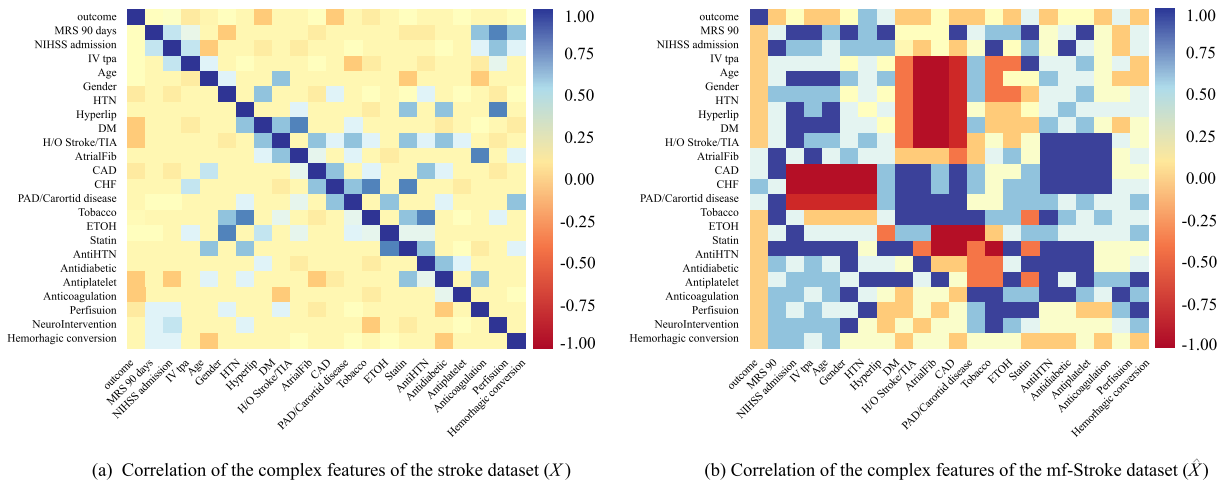


Fig. 2. Correlation depiction of the complex features in (a) stroke dataset (X) and (b) mf-stroke dataset ($\hat{X}$).

interactions are checked. In this process, multiplication and division of the non-trivial numerical columns were addressed. Thereby, 1152 columns including the original ones were generated. After dropping the highly correlated entries from the whole data (both training and test sets), the number of features has been reduced from 1152 to 23 as illustrated in Table 2. These procedures have enabled the effective handling of computational complexity, which has served the multiclassification of features in the dataset. Thus, Fig. 2 provides the correlation depiction of the complex features in the stroke dataset (X) and the mf-stroke ($\hat{X}$) dataset distinctively. As illustrated in Fig. 2 (a), it is seen that the majority of the related predictors are not correlated; whereas, in mf-stroke dataset ($\hat{X}$) (depicted in Fig. 2 (b)), it has been yielded that many variables have significant correlations between one another owing to the refining of the data and computation of singularity, self-similarity and significance by using the multifractal regularization method.

2.3. Multifractal-based AI hybrid methods

The aim of our paper is to conduct the singularity analysis for the extraction of the subtle patterns from the stroke dataset (see Table 2). Thus, the classification of three subtypes of stroke, namely small vessel, cardioembolic and cryptogenic, has been performed by the ensemble and deep-learning methods. When the complex and dynamic structures of the brain, particularly as in the case of stroke, are taken into consideration, the profound complexity of the brain is revealed. It is sometimes observed that the subtle details related to stroke could be overlooked during the regular and conventional treatment procedures, which eventually leads to adverse consequences for the patients. In the proposed mathematical model of this study, multifractal regularization method has been used to analyze singularities for the extraction of the subtle and hidden patterns in the stroke dataset, adopting two different approaches the stages of which are indicated as follows:

(i) The first approach consists of two stages: a) multifractal regularization (L2 normalization) method was applied to the stroke dataset (X) (see Table 2), as a result of which mf-stroke dataset ($\hat{X}$) was obtained. The rationale behind the application of the multifractal regularization with L2 normalization is the enabling of the calculations to be simpler since the numerical aspect is replaced by the analytical minimization, thus, distance was minimized by the L2 normalization.

(ii) In our second approach, the algorithms, deep learning methods (Feed Forward Neural Network, Multilayer Perceptron), ensemble methods (bagging methods (Random Forest and Extra Tree classifiers), boosting methods (XGBoost, Gradient Boosted, Light Gradient Boosting classifiers) and stacking method) and fundamental methods (Decision Tree, Logistic regression) and clustering methods (K-nearest neighbor, Support Vector Machine) were applied to the stroke dataset (X) and the mf-stroke dataset ($\hat{X}$). These AI algorithms assist for the earlier detection of stroke and determination of the stroke subtype, which will serve the acceleration of the treatment and improvement of the related outcomes based on the treatment by enabling decision-making in terms of prescribing accurate medication and intervention.

(iii) The comparison of the accuracy rates (along with specificity, sensitivity, precision, F1-score and Matthews Correlation Coefficient analyses) following the application of deep learning methods, ensemble methods, fundamental methods and clustering methods (ii) has been performed. Based on these applications, it is observed that the mf-stroke dataset ($\hat{X}$) yields higher accuracy rates for the classification of the three subtypes of stroke subtypes.

Consequently, the outcomes of our proposed mathematical model in this study demonstrate the importance of deep learning methods, ensemble methods, fundamental methods and clustering methods in AI along with the singularity analysis by multifractal regularization. Thus, it can be remarked that resilient advancements in AI provide the path for efficient and timely decision-making, big data analysis and so on, while at the same time play a significant role regarding the individualized treatment courses for neurological diseases like stroke. Another significance of the proposed paradigm of our study is that the multistage mathematical modeling has the capability of achieving accurate results on the different tasks, such as classification and prediction, concerning the multi-feature dataset with hidden data characterized by high degrees of complexity, which is stroke in our case.

All the analyses and the outcomes obtained based on the related analyses as depicted in the figures were performed by Matlab [83] and Python software [84] in this study.

2.3.1. Multifractal complexity analysis

Complex spatial temporal systems manifest irregular behaviors at multiple levels. When the hierarchically nested structure of complexity in temporal patterns is considered, which is also known as the hierarchical complexity, the essential nature of temporal dynamics is at stake, where patterns are observed at each of the scales. In cases when a limited number of observations is feasible, the identification and reconstruction of complex dynamical elements pose a formidable challenge. To tackle such a challenge, multifractal analysis, which examines functions, signals, images or areas through the fluctuations of their local regularity across time or space, can be effectively employed in many applications of different natures. Multifractal analysis is capable of capturing the critical attributes and properties of the related temporal and spatial dynamics.

2.3.2. Multifractal formalism and multifractal analysis for singularity analysis

Multifractal formalisms are characterized by the scale invariance of a behavior, which is defined numerically as a spectrum concerning scaling exponents particularly in heterogeneous environments and large classes of measures. The contributions by Mandelbrot upon the introduction of fractal geometry concept and fractality have led to the understanding of the complex systems dynamics in nature on both temporal and spatial scales [85]. Initially conceived based on the stochastic processes called multiplicative cascades, Yaglom (1966) and afterwards, Mandelbrot (1974) developed the multiplicative cascade to model the energy transfer at multiple scales, from the most raw scale towards the finest scales. Legendre formalism and large deviation and are two multifractal formalisms, among the certain methods for the estimation of the multifractal spectrum [86,87].

Multifractality is profoundly rooted in the inherent heterogeneous nature of the complex networks, and thereby, multifractal analysis is employed for the description of natural and physical phenomena as well as objects based on the scaling laws. Accordingly, characterized by embedded heterogeneous self-similarities and irregular patterns, fractality is ubiquitous both in the nature and also in the brain that exhibit inherent multifractal properties considering its multifaceted complex cognitive processes and hierarchal structure.

In this study, we have dealt with the refinement of complex stroke data based on the analysis of the local Hölder regularity, also known as the Lipschitz exponent. Data refinement involves the increasing of the Hölder regularity at each point [88]. The relevant method is well-aligned with the case in question, where the data to be recovered is vastly irregular. Hölder Regularity Analysis is considered to be a measure of generalization for the differentiability of a function.

The focal point in this study is concerned with the pointwise Hölder exponent to simplify the notations, whereby the data are assumed to be not differentiable [89–91].

$$\begin{aligned} & \text{Let } \alpha(0, 1) \text{ and } x_0 \in Q \subset \mathfrak{R} \text{ A function } f : Q \rightarrow \mathfrak{R} \text{ is in } L_{x_0}^\alpha \text{ if, for all } x_0, \\ & \left| f(x) - f(x_0) \right| \leq 1|x - x_0|^\alpha \end{aligned} \quad (1)$$

Here, α is a constant. Pointwise Hölder exponent of f and x_0 as expressed by $\alpha(x_0)$, is the supremum of the α . For this, (1) is valid.

The statistical approach that leads to the consideration of a quantity called the *large deviation multifractal spectrum* is another focal aspect of this study [90–92]. Accordingly, the definition of the spectrum is defined as below:

Assume a stochastic process $X(t) \ t \in T \subset \mathbb{R}$ across a probability space (Ω, F, P) . For notation convenience, we shall reckon without loss of generality as $T = [0, 1]$. Set $N_n^\varepsilon(\alpha) = \#q : \alpha - \varepsilon \leq \alpha_n^q \leq \alpha + \varepsilon$, in which α_n^q is the coarse-grained Hölder exponent, corresponding to the dyadic interval $I_n^q = [q2^{-n}, (q+1)2^{-n}]$; that is $\alpha_n^q = \log |Y_n^q| / -\log n$. In this regard, $(\#)$ refers to the number of nonempty boxes, and the related measure is conveyed by exponents from the interval $(\alpha - \varepsilon, \alpha + \varepsilon)$ [90]. Y_n^q is a quantity that measures the variation of (X) in the interval I_n^q . The option $Y_n^q := X((q+1)2^{-n}) - X(q2^{-n})$ results in more simple computations related to analytical aspects. Another possibility, in our study, would be to take Y_n^q as the $x_{n,q}$ of (X) at scale n and location q [90]. This definition is expedient in different ways as it enables the employment of the versatility regarding the wavelet bases. Yet, this would also have a drawback. The multifractal spectrum obtained in this manner will principally rely on the chosen wavelet ψ . Thus, if $Y_n^q : x_{n,q}$ is set like this, it is not reasonable to address of the spectrum of (X) without having the reference of analyzing the chosen wavelet. Through the manuscript, we denote the wavelet coefficient of data x by $x_{j,q}$ with j which is the scale and q the location [90–92]. Subsequently, the coarse-grained multifractal large deviation spectrum is obtained by the equation spectrum, which can be provided by this equation $f_g(\alpha) = \lim_{\varepsilon \rightarrow 0} \lim_{n \rightarrow \infty} \sup (\log N_n^\varepsilon / \log n)$. The $f_g(\alpha)$ definition is associated with the large deviation theorem which has enabled the provision of the probabilistic interpretation concerning the multifractal spectrum.

It can be stated that in the choice of Y_n^q, f_g the time ranges in $\mathbb{R}^+ \cup \{-\infty\}$. Here, $-\infty$ corresponds to the values belonging to the coarse-grained exponent, and these are not adequately observed on a small scale [90]. The heuristic meaning of f_g is to be defined as follows: For n that is adequately large, one has approximately $P_n(\alpha_n^q \simeq \alpha) \simeq 2^{-n(1-f_g(\alpha))}$ where P_n gives the uniform distribution over $\{0, 1, 2, \dots, 2^n - 1\}$. Hence, for all α such that $f_g < 1, 1 - f_g(\alpha)$ makes the measurement of the exponential rate of decay for the probability of finding an interval I_{qn} which has a coarse-grained exponent equaling to α , while n leans towards infinity. f_g , as a whole, is a random function. During the applications, it is expedient to consider the deterministic version of f_g as follows [90]:

$$F_g(\alpha) = 1 + \lim_{\varepsilon \rightarrow 0} \lim_{n \rightarrow \infty} \sup \frac{\log \pi_n^\varepsilon(\alpha)}{\log(n)} \quad (2)$$

$$\text{where } \pi_n^\varepsilon(\alpha) := P \times P_n[\alpha_n^q \in (\alpha - \varepsilon, \alpha + \varepsilon)].$$

In this section, we present the technique which is reliant on the multifractal data rather than making the application of it merely for the use of the Hölder exponent. Such an approach can often assure more robust estimations since a higher level description is employed to subsume information along the entirety of the data. Furthermore, a semiparametric approach has also been assumed, which means that we assume that the data at stake belong to a particular set of parameterized classes, as elaborated herein [93,94]. F refers to the set of lower semicontinuous functions from $\mathbb{R}^+$ to $\mathbb{R}^+ \cup \{-\infty\}$ [90]. The random functions' classes are regarded as $X(t), t \in [0, 1]$. This is defined as (Ω, F, P) and in line with Definition 1 provided below. All clusters $S(g, \psi)$ are taken as per the functional parameter $g \in F$ and the wavelet ψ . In addition, the set $\{x_{j,q}\}$ makes up a basis of L_2 . Q , here, is a positive constant, whose definition is in line with the following equation:

$$P_j^\varepsilon(\alpha, Q) = P \times P_j \left(\alpha - \varepsilon < \frac{\log_2(Q|x_{j,q}|)}{-j} < \alpha + \varepsilon \right) \quad (3)$$

Definition 1 is $S(g, \psi) = \{X : \exists Q > 0, j_0 \cup Z : \forall j > j_0, x_{j,q}, x_{j,d}\}$ and its distribution is done in an identical way. $(q, q') \in \{0, 1, \dots, 2^j - 1\}$ and $\log_2 P_j^\varepsilon(\alpha, Q)/j = g(\alpha) + \mathfrak{R}_{n,\varepsilon}(\alpha)$ is $\lim_{\varepsilon \rightarrow 0} \lim_{n \rightarrow \infty} \mathfrak{R}_{n,\varepsilon}(\alpha) = 0$ uniformly in α .

For j that is adequately large, the wavelet coefficients $(x_{j,q})_q$ at scale j are assumed to be distributed in an identical way as follows:

$$\pi_n^\varepsilon[\alpha] := P \times P_n[\alpha_n^q \in (\alpha - \varepsilon, \alpha + \varepsilon)],$$

$$\pi_n^\varepsilon[\alpha, Q] := P \times P_j \left(\alpha - \varepsilon < \frac{\log_2(Q|x_{j,q}|)}{-j} < \alpha + \varepsilon \right)$$

$$= P \left(\alpha - \varepsilon < \frac{\log_2(Q|x_{j,q}|)}{-j} < \alpha + \varepsilon \right) \quad (4)$$

As a result of these applications, Definition provides a plain interpretation on multifractal analysis. We consider the set of random data (X) for a given wavelet ψ . We consider the set of random data X for a given wavelet ψ . Thereby, the normalized data QX owns the deterministic multifractal data $Fg(\alpha)$ equaling to $1 + g$ besides this next condition: F_g is obtained as a limit in $j \lim \sup$. This limit was obtained in *uniformly* into α , verifying the rescaled statistics of the $\alpha_{j,q}$, which is close enough to their limit for large j , ensuing an inference that is significant. The cluster $S(g, \psi)$ depicts a broad array of data.

Herein, the principal steps in the classical *Maximum A Posteriori* (MAP) approach in a Bayesian frame is known to be adjusted to our case. The noisy data is Y , and we presume that $Y = X + B$. B denotes noise which is not dependent on the original data X as per the rule as B (orthonormal wavelets have been used here, where B is stated in the related equation as b). Thus, we get $y_{i,j} = x_{j,q} + b_{j,q}$. The map estimate $\hat{x}_{j,q}$ of $x_{j,q}$ from the observation $y_{j,q}$ is defined as an argument maximizing $P(x_{j,q}/y_{j,q})$. As $P(y_{j,q})$ is not dependent upon $x_{j,q}$, employing Bayes rules and maximizing $P(x_{j,q}/y_{j,q})$ signify maximizing the product $P(x_{j,q}/y_{j,q})P(y_{j,q})$.

The MAP estimate is stated as $\hat{x}_{j,q} = \arg \max_x [P(y_{j,q}/x)P(x)]$. $P(x_{j,q}/x)$ can be computed as based on the rule of B easily if B is taken to be (0) [92]. This is because $b_{j,q}$ share the same rule as B . The preceding $P(x_{j,q})$ is drawn based on our assumption that X belongs to $S(g, \psi)$ indicated as in Equation 5: for $x > 0$, set $\alpha_j(x) = \log_2(Qx)/j$,

$$P(|x_{j,q}| = x) = P\left(\frac{\log_2(Q|x_{j,q}|)}{-j} = \alpha_j(x)\right) \simeq 2^{j(g(\alpha_j(x))^{-1})} \quad (5)$$

This gives the definition of an approximate Bayesian MAP estimate as in equation 6:

$$\hat{x}_{j,q} = \arg \max_{x>0} \left[jg\left(\frac{\log_2(\hat{Q}x)}{-j} - \frac{(y_{j,q} - x)^2}{2\sigma^2}\right) \right] \cdot \text{sgn}(y_{j,q}) \quad (6)$$

where $\text{sgn}(y)$ shows the sign of y and $\hat{Q} = \left(\sup_{j>j_0} \sup(x_{j,q})\right)^{-1}$. The estimate for Q can be justified heuristically as in: $\log_2(Q|x_{j,q}|) / -j \simeq \alpha$ with $\alpha > 0$ indicating that $Q|x_{j,q}| < 1$ for all the couples (j, q) . $\hat{Q}$ is selected as the smallest normalizing factor which requires the inequality indicated subsequently. In the experiments of our study, we deal with the case where the noise is centered, Gaussian, with variance σ^2 . The MAP estimate can be indicated as per Equation 7;

$$\hat{x}_{j,q} = \arg \max_{x>0} \left[jg\left(\frac{\log_2(<\hat{Q}x)}{-j} - \frac{(y_{j,q} - x)^2}{2\sigma^2}\right) \right] \cdot \text{sgn}(y_{j,q}) \quad (7)$$

Equation (6) presents an evident formula for Y denoising, but it ensures a limited use in terms of practicality. As a matter of fact, it is not always possible to be aware of the multifractal data of X in many of the applications. If there does not exist any of g , it would not be likely to use (6) to obtain $\hat{x}_{j,q}$. Additionally, F_g in general, is dependent on the analyzing wavelet so it is required that one comprehends the shape of the data for the particular wavelet. The principal goal of our approach is to identify the multifractal characteristics of X from the stroke data $\hat{X}$: the Multifractal Bayesian approach in this study provides the estimation of F_g^x as follows: (a) denoise Y , (b) evaluate the data $\hat{F}_g^x$ numerically, (c) set $\hat{F}_g^x = \hat{F}_g^x$. It is evident based on this approach that previous knowledge of $\hat{F}_g^x$ in the Bayesian approach is not always required. We have, therefore, presented a “degenerated” version of (6), where the input is taken as a single real parameter, instead of addressing the entire data. The heuristic is based on regularity perspective, as important information in the data, based on its support, for instance, the set of all the Hölder exponents. α_0 signifies the smallest regularity which is actually observed in the data [95,96]. The shapes of the F_g spectra which is as attained by different analyzing wavelets depend upon the wavelet, and yet, their support is contained in $[\alpha_0, \infty)$. The “flat data”, for this reason, includes the inherent information. It is based upon the positive real α_0 . The re-expression of (6) with a flat data provides the expression as in Equation 8:

$$\hat{x}_{j,q} = \begin{cases} y_{j,q} & \text{if } |y_{j,q}| < 2^{-j\alpha_0} \\ 2^{-j\alpha_0} & \text{otherwise.} \end{cases} \quad (8)$$

α_0 is really a priori information; yet, it is possible to predict it from the noisy observations. Given that fact, it bears analogy to the threshold employed in the classical hard or soft wavelet thresholding scheme. Regarding α_0 as a tuning parameter in the applications will prove to be useful. Having α_0 increased provides a *smoother estimate* as well; this is because we assume that the stroke data have a minimal exponent that is larger [90].

2.3.3. Numerical experiments based on multifractal analysis and singularity analysis

Multifractal analysis, as an efficient method to recovering irregular and heterogeneous stroke data (see Table 2 and Table 3) is used in our multifractal scheme. The analysis of the pointwise structure of a singular signal is carried out by multifractal spectrum, which yields probabilistic or geometrical information concerning the distribution of points which bear the same singularity. The previous characterization is gained by the Hausdorff spectrum f_h , which the latter one presented by the large deviation spectrum f_g , and $f_h \leq f_g$ refers to the Hausdorff dimension of the points [49,97]. Therein, the Hölder exponent equals to α , whereas $f_g(\alpha)$ makes the evaluation of the convergence speed pertaining to the probability, with the coarse-grained exponents at resolution ε equaling to α as ε tends toward 0. In some cases, f_g and f_h overlap with each other. The phenomenon provides a means for the evaluation of f_h , it also enables the drawing of a number of conclusions for the practical matters concerning the signal.

Concerning f_h , in the one-dimensional case, the Hölder exponent of signal f at x_0 is as follows:

$$\alpha(x_0) = \sup \left\{ \tilde{\alpha} : \exists P_m, \exists C > 0 / f(x) - P_m(x - x_0) < C|x - x_0|^{\tilde{\alpha}} \right\} \quad (9)$$

P_m denotes a polynomial of degree that does not exceed the integer part of $\tilde{\alpha}$ at x is located in the neighborhood of x_0 . The Hausdorff multifractal spectrum is shown in the following way:

$$f_h[\alpha] = \dim_H E(\alpha) = \dim_H \{x | \alpha(x) = \alpha\} \quad (10)$$

where $\dim_H$ signifies the Hausdorff dimension [98].

The stroke dataset's multifractal analysis includes the computing of its multifractal spectrum as well as the classification of each point based on the corresponding value of $(\alpha, f_h(\alpha))$ geometrically. The α value offers the local information regarding the pointwise regularity, and the $f_h(\alpha)$ value provides global information.

The results of numerical experiments pertaining to the stroke dataset are presented, and the outcome of the Multifractal method (L2 normalization) is presented in all the related cases. The attributes (see Table 1 and Table 2) are set in such way to best fit to the known stroke data. A far-reaching array of sampled functions of length are addressed to estimate the local regularity for each stroke attributes (see Table 2). Numerical experiments, in our proposed approach, were obtained by Multifractal method (L2 normalization) (see Table 2). The multifractal method [99] was applied on the stroke dataset so that we could obtain the self-similar, irregular and significant, stroke datasets (mf-stroke dataset ($\hat{X}$)). The subsequent steps have been applied to this end:

The attributes of the stroke dataset are represented as follows: signal $s(x, y)$, noise $n(x, y)$, observed signal $w(x, y)$, and $z(x, y)$ refers to the estimate of the signal, and (x, y) is denoted as the pixel location. For the recovering of the images, a regularized version $z(x, y)$ of $w(x, y)$

Step 1. $z(x, y)$ is close to $w(x, y)$ in the L^2 sense, indicating that $\|z(x, y) - w(x, y)\|_{L^2}$ is minimum. The framework for regularization involves that we have no assumption concerning a likely noise. Here, a signal Z is sought and this is close in the L^2 sense to the observations Y (our input signal), and with regularity that is equal to that of Y and a shift. The typical values for the shift in this case also approximate towards 2.

Step 2. The local Hölder function of ($\hat{X}$) is depicted. Let $\alpha_z = \alpha_w + \delta$ where $\alpha_z = \alpha_s$ refers to the Hölder regularity of the estimated signal. δ is a positive function which is defined by the user. In this way, the z regularity is going to be everywhere larger compared to the one of the observations. The estimation of the Hölder exponent is reliant on the wavelet procedure.

Have $\{\psi_{j,k}\}$ as the orthonormal wavelet basis in which j signifies scale and k signifies position. Considering $\{\psi_{j,k}\}$ has sufficient number of vanishing moments, wavelet coefficients behave as $g_k(k) \approx 2^{j(\alpha+1/2)}$; this is because j tends to $-\infty$. α , here, shows the Hölder exponent, rendering the quantification of local variation, obtained from the regression of the logarithm of the wavelet coefficients of $z(x, y)$ at any above point i related to the scale, which is: $-(\alpha(i) + 1/2)$. $\alpha_z = \alpha_w + \delta$, where $\alpha_z = \alpha_s$ signifies the estimated signal's Hölder regularity, δ denotes a positive function which is defined by the user. For this reason, the regularity will possess a higher value than one of the observations everywhere. The Hölder exponent is also estimated depending on a wavelet-based procedure [81,100,101]. In this study, multifractal regularization (L2 normalization) technique was applied to the stroke dataset (X). After multifractal regularization was applied to the stroke dataset, the mf-stroke dataset ($\hat{X}$) was generated as consequence of regularity based enhancement ($\hat{X}$) from multifractal regularization method.

2.3.4. Ensemble predictive methods

Having the capability of handling non-linear, complex, non-stationary and multidimensional data, an ensemble method, as a technique related to machine learning, performs the combination of different base models for the generation of one optimal predictive model [102]. Ensemble learning is a paradigm of machine learning which multiple models are trained for the solution of the same problem, with a combination obtained so that more accurate results can be yielded. Since the results are yielded more accurate compared to the results of individual classifiers, ensemble methods are favored [103–106]. An ensemble of classifiers is able to learn a more complex decision in a more accurate way, making the splitting of the different classes efficiently [107–109]. Moreover, like the ensemble methods, multiple classifier systems are capable of extracting more significant information from the raw data.

As addressed in the current study, the prediction of the subtypes of stroke includes a task related to classification processes, which requires function f that is a linear combination of the explanatory variables (features) of a given observation with coefficients using a dot product:

$$f(x_i, k) = x_i \cdot \beta_k \quad (11)$$

where x_i is the vector of predictor variables defining observation i , β_k is a vector of weights (or coefficients) corresponding to outcome k , and $f(x_i, k)$ is the probability (or score) associated with selecting observation i to class k . As this study involves a labeled data set, the relevant classification task belongs to a supervised learning problem [110].

Ensemble learning is an approach in which multiple models are employed to solve a particular computational complexity, and ensemble techniques combine different machine learning models with one particular ensemble predictive model. In most cases, ensemble methods are mainly employed to enhance a model's overall performance. When compared to using one particular model to solve a problem, the ensemble models merge several different models to predict a final model's results. For instance, the bias-variance trade-off is the essential benchmark that distinguishes a robust model from a poor one. Bias is known to be a failure that occurs due to false assumptions in the training algorithm, which is the pillar of any learning pattern. In machine learning, the models with a high bias are inclined to have a lower variance, so high bias can make a machine-learning algorithm skip important information and eventually miss direct correlations between independent variables and the class labels, which causes the model to underfit. High variation in a machine learning algorithm will make the model inclined toward random noise in the dataset, which brings about the overfitted model.

There exist basically two approaches for the combination of models in ensemble predictive modeling. On the one hand, homogeneous ensemble has all the classifiers being of the same type; whereas, heterogeneous ensemble has different classifiers. The reason for using heterogeneous methods is owing to the fact that different models present different views regarding the data at hand, and they also rely on different mathematical paradigms. The model proposed by our study falls into the heterogeneous category whose advantage lies explicitly in capturing the dependencies between the classifier predictions. The algorithms in this category are tested

against different benchmark datasets, and the results obtained show that heterogeneous approaches yield more accuracy, stability, effectiveness and efficiency.

In the current study, among the common deep learning methods (Feed Forward Neural Network, Multilayer Perceptron), three of the well-known heterogeneous ensemble methods (bagging methods (Random Forest and Extra Tree classifiers), boosting methods (XGBoost, Gradient Boosted, Light Gradient Boosting classifiers) and stacking method) and fundamental methods (Decision Tree, Logistic regression) as well as clustering methods (K-nearest neighbor, Support Vector Machine) have been addressed. In the proposed model of our study, the stacking method has been compared with methods which are Decision Tree, Extra Tree Classifier, Feed Forward Neural Network, Gradient Boosting algorithm, K-nearest neighbor, Light gradient boosting machine, Logistic regression, Multilayer Perceptron, Random Forest, Support Vector Machine and Extreme Gradient boosting. The experimental results obtained from our study validate the fact that the heterogeneous stacking ensemble technique is superior to the paradigms mentioned above applied in the paper to detect the stroke subtypes specified.

2.3.5. Bagging method

As an ensemble method, bagging, has been proposed based on bootstrapping and aggregating methods, which are implemented frequently to reduce the variance in a given model. The preliminary benefit of this method is to get the noise reduction in the observations existent in random samples [111,112]. Another advantage of bagging is concerned with the improvement of classification and regression performance while reducing the variance compared to the base procedure. Algorithm 1 provides the application steps concerned with the bagging algorithm for classification purposes.

Algorithm 1 Bagging algorithm for classification.

Input: · Base learning algorithm B .

· Number of bootstrapped samples τ .

· Training data D is the set of m training examples.i.e. $D = \{x_i, y_i\}_{i=1}^m (x_i \in R^m, y_i \in Y)$.

Output: An ensemble classifier C^* .

1: **for** i from 1 to τ **do**

2: Step 1.1: Create a bootstrapped training set S of size n .

3: This S is a samples with replacement from D .

4: Create a new classified from $C_i = B(S)$ by learning a specific base learner $B(x, y)$ on S .

5: **end for**

6: Finally create a classifier

$$C^* = \arg \max_y \sum_{i=1}^{\tau} 1_{\{C_i(x)=y\}}$$

7: **Return** C^*

Concerning the stroke dataset handled in this study, we assume our base learner as Random Forest and 200 as a number of bootstrapped samples. To this end, 200 different bootstrap subsamples of the training data for stroke dataset (X) and mf-stroke dataset ($\hat{X}$) have been obtained. We performed the training of the random forest, indeed, constructing trees through the random selection of features based on these distinct 200 samples. Subsequently, we have employed the majority vote in order to obtain the final outcomes.

2.3.6. Boosting method

Boosting, as a parallel ensemble method, is able to convert a weak learner into a learner which has a higher accuracy level. As is the case with bagging, in boosting, an iterative learning process is implemented. Thus, one distinctive characteristic of boosting is that each iteration employs a reweighting system to be able to focus on the learning process, where the observations in the previous steps were estimated with more errors [111,112]. One difference between bagging is that boosting methods are considered to be sequential ensemble algorithms in which the training of the weak learners is done iteratively. In this way, the error in the former mode is attempted to be corrected. Algorithm 2 provides the application steps concerned with the boosting algorithm for classification purposes.

Note the dimensions of training datasets for stroke dataset and mf-stroke dataset is (1156×23) . We tried different number of iterations for different boosting algorithms, for example, 5000 iterations for extreme gradient boosting algorithm while 7000 iterations for light gradient boosting machine. We used multi-class cross-entropy loss functions as our differentiable loss function. Initially, in order not to be biased towards, a constant function γ other than 0 was performed. Then, $\sum_{i=1}^{5000} L(y_i, \gamma)$ when L is multi-class cross-entropy loss function. To compute the pseudo-residuals, derivatives have been obtained, and following the fitting of the base models to the modified datasets, the process 5000 has been iterated for the final output.

2.3.7. Stacking generalization method

Stacking methods involve the process of learning an ensemble of models. The outputs of this process have the function of being meta-features to a meta-mode. Stacking is regarded as a heuristic way to be able to enhance the performances of the weak learners by

Algorithm 2 Boosting algorithm for classification.**Input:** · Differentiable loss function $L(y, F(x))$.· Number of iterations M as an input.· Training data D is the set of m training examples, i.e. $D = \{x_i, y_i\}_{i=1}^m (x_i \in R^m, y_i \in Y)$.**Output:** An ensemble classifier C^* .

1: Step 1: Initialize the model with a constant value:

$$F_0(x) = \arg \min_{\gamma} \sum_{i=1}^m L(y_i, \gamma).$$

2: **for** i from 1 to M **do**

3: Compute pseudo-residuals

$$r_{im} = - \left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \right]_{F(x) = F_{m-1}(x)} \text{ for } i = 1, \dots, n$$

4: Fit a base learner $h_m(x)$ to pseudo-residuals, i.e. train using the training set $\{(x_i, r_{im})\}_{i=1}^n$.5: Compute multiplier γ_m by solving the following one-dimensional optimization problem

$$\gamma_m = \arg \min_{\gamma} \sum_{i=1}^n L(y_i, F_{m-1}(x_i) + \gamma h_m(x_i))$$

6: Update the model:

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x).$$

7: **end for**8: **return** Output $F_M(x)$

considering the learners' errors and their correlations [113]. In terms of its predictive power and generalization capability, the stacked generalization ensemble technique has superiority when compared with the aforementioned methods. A stacked generalization yields a nonlinear and complex model, which can be employed to make the combination of weak learners, converting them into most possible rigorous model. Two phases are involved in stacking. Initially, the training of the different base classifiers is done on the training data and in the second phase, the output of the first stage functions as an input. In other words, if there is a misclassification by the classifier and this error is detected from the data from a specific place, then this behavior is to be learned by the supervisor model. The error is corrected accordingly [114,115].

The stacking generalization algorithm with k-fold cross validation illustrates the related steps in this process as depicted in Fig. 3 regarding Algorithm 3.

Algorithm 3 Stacking generalization algorithm with k-fold cross validation.**Require:** Decomposition of data into two sets as training and test sets.**Input:** Training Data $D = \{x_i, y_i\}_{i=1}^m (x_i \in R^m, y_i \in Y)$ **Output:** An ensemble Classified H

1: Step 1: Adopt k-fold cross validation approach in preparing training data set for level-1 classifier.

2: Divide the data set D into equally k-fold subsets $\{D_1, D_2, \dots, D_K\}$ 3: **for** i from 1 to K **do**

4: Step 1.1: Learn level-0 classifiers.

5: for j from 1 to L **do**6: Learn a base classified h_{ij} from D/D_i 7: **end for**

8: Step 1.2: Construct a training set for level-1 classifier.

9: **for** $x_i \in D_k$ **do**10: Get a result $\{x'_i, y_i\}$, where $x'_i = \{h_{k1}(x'_i), h_{k2}(x'_i), \dots, h_{hL}(x'_i)\}$ 11: **end for**12: **end for**

13: Step 2: Learn a level-1 classifier.

14: We call the collection of $\{x'_i, y_i\}$ new level-0 data.15: Train the meta learning algorithm h on a newly constructed level-0 data set.

16: Step 3: Re-learn level-0 classifiers

17: **for** j from 1 to L **do**18: Learn a classifier h_j based on D 19: **end for**20: **return** $H(x) = h'(h_1(x), h_2(x), \dots, h_L(x))$

The steps related to the proposed stacking method in Algorithm 3 are shown to be designed to deal with learning a meta-level classifier to join multiple weak-level classifiers' predictions. The rigor of the stacking method stems from its strength to utilize the

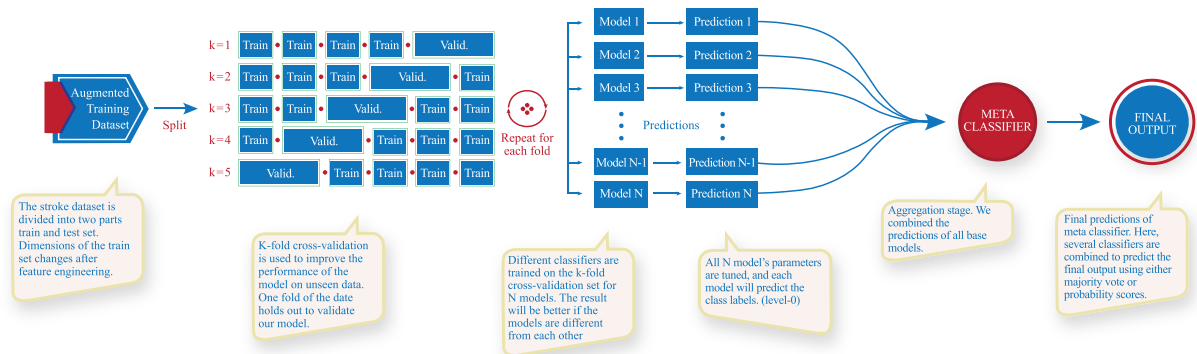


Fig. 3. Design of stacked generalization ensemble model to construct the highest predictive power.

diversity in base-level classifiers' predictions, thus predicting higher precision at the meta-level. Accordingly, Fig. 3 illustrates the design of stacked generalization ensemble model to construct the highest predictive power.

In the augmented training set, we fixed the imbalanced outcome by resampling it. The first step is to combine the train and validation set. The second step is balanced by the resampling technique.

Thus, as a result of the technological and computing advances as well as AI, mathematical models along with forecasting methods have become more precise, accurate and sophisticated. In this study, Multilayer Perceptron, Light Gradient Boosting machine, and Stacking generalization algorithms are utilized on the stroke dataset (X) and mf-stroke dataset ($\hat{X}$) for prediction purposes. Through these models, forecasting operations can be performed efficiently owing to their mapping capabilities and learning ability. Predicting the accurate stroke subtype is vital in healthcare, for the patient outcomes including life-quality, (self-)control and survival. Given these aspects, predictive models are critical for the accurate classification, evaluation, multicriteria decision-making, estimation and prediction to address the issue of uncertainty, variability and unpredictability.

3. Experimental results and discussion: the application and comparative optimized predictive performance of multifractal analysis, ensemble and AI methods

This study has employed deep learning, deep learning methods (Feed Forward Neural Network, Multilayer Perceptron), ensemble methods (bagging methods (Random Forest and Extra Tree classifiers), boosting methods (XGBoost, Gradient Boosted, Light Gradient Boosting classifiers) and stacking method) and fundamental methods (Decision Tree, Logistic regression) as well as clustering methods (K-nearest neighbor, Support Vector Machine) along with multifractal analysis within a multistage combination scheme for multifeatured complex stroke dataset (X). In line with our proposed approach and mathematical model, deep learning and ensemble methods in AI along with the singularity analysis have been utilized for the extraction of subtle patterns in the stroke dataset (X) by the multifractal regularization method. By the utilization of the multifractal regularization method on the stroke dataset, we have been able to generate the mf-stroke dataset ($\hat{X}$). Afterwards, the algorithms consisting of the ensemble methods and deep learning methods have been applied both to the stroke dataset (X) and the mf-stroke dataset ($\hat{X}$). Last but not least, we have compared the performance accuracy rates, considering the specificity, sensitivity, precision, F1-score and Matthews Correlation Coefficient analyses. Hence, comparative prediction performances obtained as a result of the above steps have been conducted for the stroke three subtypes' classification. Accordingly, Table 4 presents the outline of the related steps belonging to our integrated multistage mathematical model proposed in the current study.

The experimental results of our proposed model reveal the importance of deep learning methods (Feed Forward Neural Network, Multilayer Perceptron), ensemble methods (bagging methods (Random Forest and Extra Tree classifiers), boosting methods (XGBoost, Gradient Boosted, Light Gradient Boosting classifiers) and stacking method) and fundamental methods (Decision Tree, Logistic regression) as well as clustering methods (K-nearest neighbor, Support Vector Machine) along with the singularity analysis by multifractal regularization. The modeling paradigm proposed by this study with its multistage form can enable the achieving of accurate results on different tasks, such as classification and prediction, concerning the multi-feature dataset with hidden data characterized by high degrees of complexity, which is stroke in our case. The absence of earlier and previous knowledge causes the clustering analysis to pose a challenging concern though many clustering algorithms. It is also acknowledged that one single clustering algorithm cannot tackle all types of distribution of data efficiently. Given that concern and constraint, each clustering algorithm has its own particular strategy so that a structure from a dataset can be discovered. Concerning the proposed integrated modeling paradigm in this study, the schemes that bring about the experimental results, as outlined in Table 4, are schematized below in their corresponding order:

Step 1: The Application of Multifractal Analysis to the stroke dataset (X), and the Generation of mf- stroke dataset ($\hat{X}$).

Multifractal analysis enables the characterization of the uneven distribution with quantitative indicators like heterogeneity, self-similarity, regularity and singularity, which helps the better revealing of the causes of changes and complexity on multifractal spectrum. To serve this purpose, one of the multifractal methods, namely multifractal regularization method (L2 normalization) is applied

Table 4

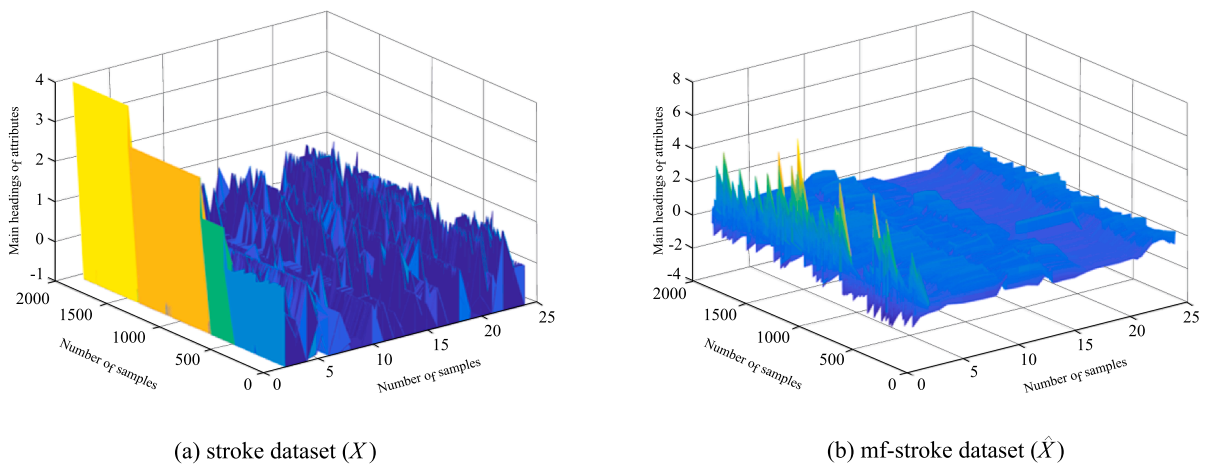
Outline of the steps followed in the integrated multistage approach.

Step 1: The application of multifractal analysis to the stroke dataset (X), and the generation of mf-stroke dataset ($\hat{X}$). (Fig. 3 displays the stroke dataset (X) and mf-stroke dataset ($\hat{X}$) on the Mesh plots.)

Step 2: PCA and its Performance evaluation with different scalers on stroke dataset (X) and mf-stroke dataset ($\hat{X}$)

Step 3: The application of optimized ensemble predictive modelling: Bagging, Boosting and Stacking generalization methods

Step 4: The comparative optimized prediction performances with accuracy rate results based on deep learning methods (Feed Forward Neural Network, Multilayer Perceptron), ensemble methods (bagging methods (Random Forest and Extra Tree classifiers), boosting methods (XGBoost, Gradient Boosted, Light Gradient Boosting classifiers) and stacking method) and fundamental methods (Decision Tree, Logistic regression) and clustering methods (K-nearest neighbor, Support Vector Machine) for the stroke subtypes' classification related to the stroke dataset (X) and mf-stroke dataset ($\hat{X}$)

**Fig. 4.** Contour plot depiction under mesh plot analysis for (a) stroke dataset (X) and for (b) mf-stroke dataset ($\hat{X}$).

to the stroke dataset (X) in our study. Thus, the generation of the mf-stroke dataset ($\hat{X}$) has been realized to detect multifractal properties such as heterogeneity, self-similarity, singularity and regularity.

Fig. 4 schematizes the stroke dataset (X) and mf-stroke dataset ($\hat{X}$) on the Mesh plots. Fig. 4(a) presents the main attributes of the stroke dataset (X), and Fig. 4(b) is concerned with those of mf-stroke dataset ($\hat{X}$), following the application of the multifractal regularization (L2 normalization) to the stroke dataset (X).

Step 2: PCA and Its Performance Evaluation with Different Scalers on stroke dataset (X) and mf-stroke dataset ($\hat{X}$) Principal component analysis (PCA) is a non-parametric data analysis process which involves the computation of principal components. It is employed to conduct a change based on the data so that useful information from complex datasets can be obtained to make predictive models. PCA provides the description of the new features by reducing dimensions for some data to be able to find the hidden or simplified structures while preserving the variation of the data as possible. In this study, the stroke dataset (X) handled contains categorical or binary variables and only three continuous variables. We have employed the PCA technique to convert all these binary or categorical variables into continuous variables. Thus, the various scalers have been compared before applying the PCA transformation to see the effects of scalers.

Fig. 5 presents the boxplots of the five main PCA components (no transformer, standard scaler, quantile transformer, robust scaler, normalizer, maximum absolute scaler, minimum maximum scaler, power transformer) based on the outcome labels where each of the 23 attributes of the three stroke subtypes' box plots concerns the stroke dataset (X) and mf-stroke dataset ($\hat{X}$). Following the application of PCA, all the steps (i-viii) on the first component without the scalers are shown in the first row. The first component attributes as "H/O Stroke /TIA" are based on the results obtained from the stroke dataset (X)

Fig. 6, on the other hand, provides the depiction of the first and second principal components of eight different PCAs for the (a) stroke dataset (X) and (b) mf-stroke dataset ($\hat{X}$) with the following details: (i) PCA with a Power transformer scaler, (ii) PCA with a maximum scaler, (iii) PCA with a Maximum Absolute scaler, (iv) PCA with Normalizer scaler, (v) PCA with Robust scaler, (vi) PCA with Quantile Transformer scaler, (vii) PCA with a standard scaler and (viii) PCA with no Transformer scaler. In a robust scaler, it seems there are many outliers in first main component. Except the robust scaler and normalizer scaler, we can see the trend of boxes becomes smaller while making a transition from the first main component (the most informational) to the fifth one (the least

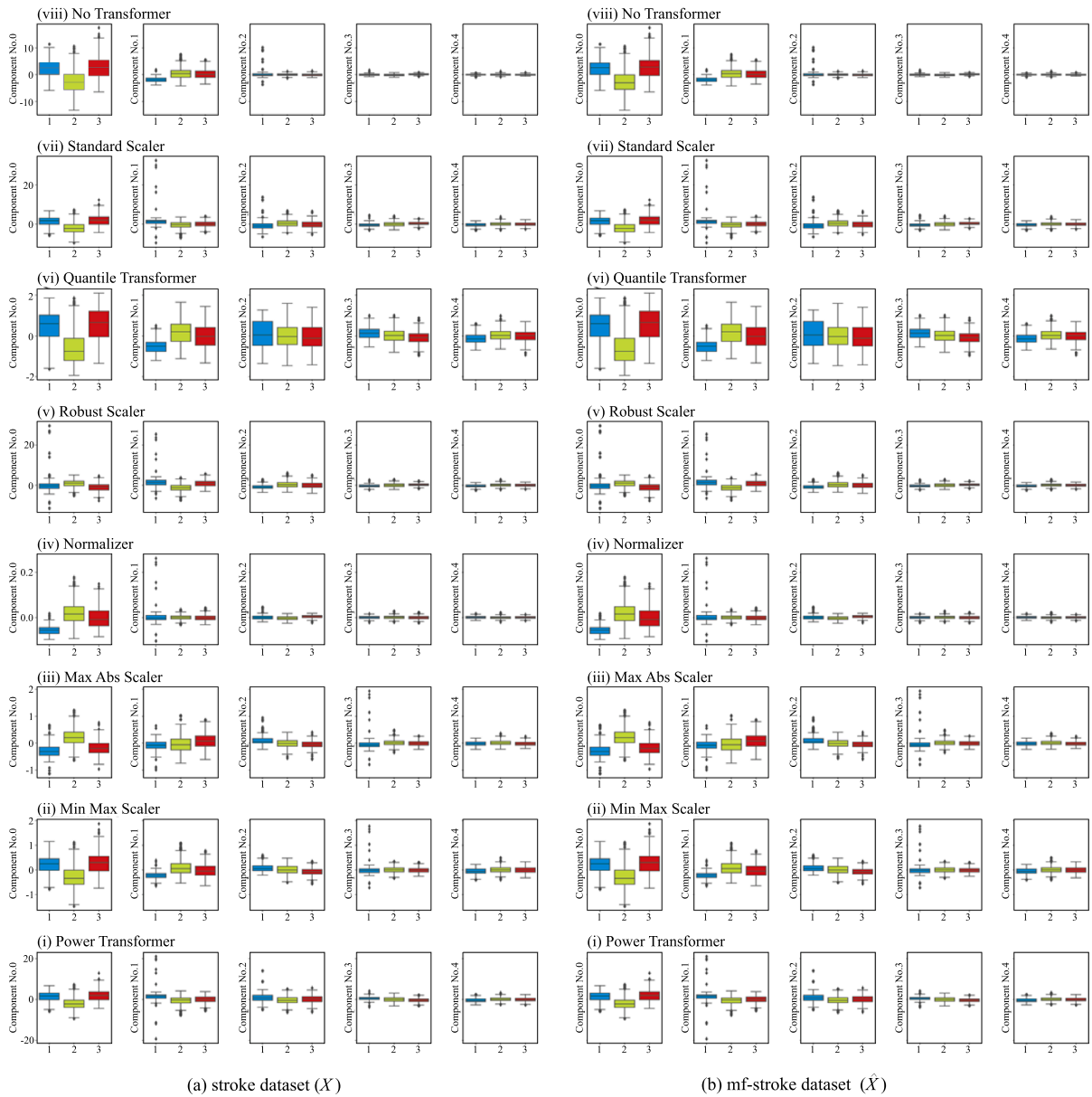


Fig. 5. The boxplots of the PCA components for (a) stroke dataset (X) and (b) mf-stroke dataset ($\hat{X}$) for (i) Power Transformer (ii) Minimum Maximum Scaler (iii) Maximum Absolute Scaler (iv) Normalizer (v) Robust Scaler (vi) Quantile Transformer (vii) Standard Scaler (viii) No Transformer.

informative). It should be noted that a cut-off effect at the fifth component by looking explained variance has been utilized to be able to retrieve the most informative and significant features.

In Fig. 7, we showed and explained the variance of various scalers on the stroke dataset and mf-stroke dataset. One can notice that when there is no transformer or Normalizer scaler, only one variable explains much variance that shows that one variable is distributed in more extensive range. Before applying PCA, one needs to check the variance of each feature in the dataset. It is also important to scale the data with a plausible scaler if there is a considerable gap between the variances.

Step 3: The Application of Ensemble Predictive Modeling: Bagging, Boosting and Stacking Generalization Methods

In the integrated multistage mathematical model proposed, deep learning methods (Feed Forward Neural Network, Multilayer Perceptron), ensemble methods (bagging methods (Random Forest and Extra Tree classifiers), boosting methods (XG-Boost, Gradient Boosted, Light Gradient Boosting classifiers) and stacking method) and fundamental methods (Decision Tree, Logistic regression) as well as clustering methods (K-nearest neighbor, Support Vector Machine) algorithm have been applied to the stroke dataset (X) and mf-stroke dataset ($\hat{X}$) for the classification and diagnosis of three different stroke subtypes (small vessel, cardioembolic and cryptogenic stroke). Table 7 presents the optimized prediction results obtained for the applications.

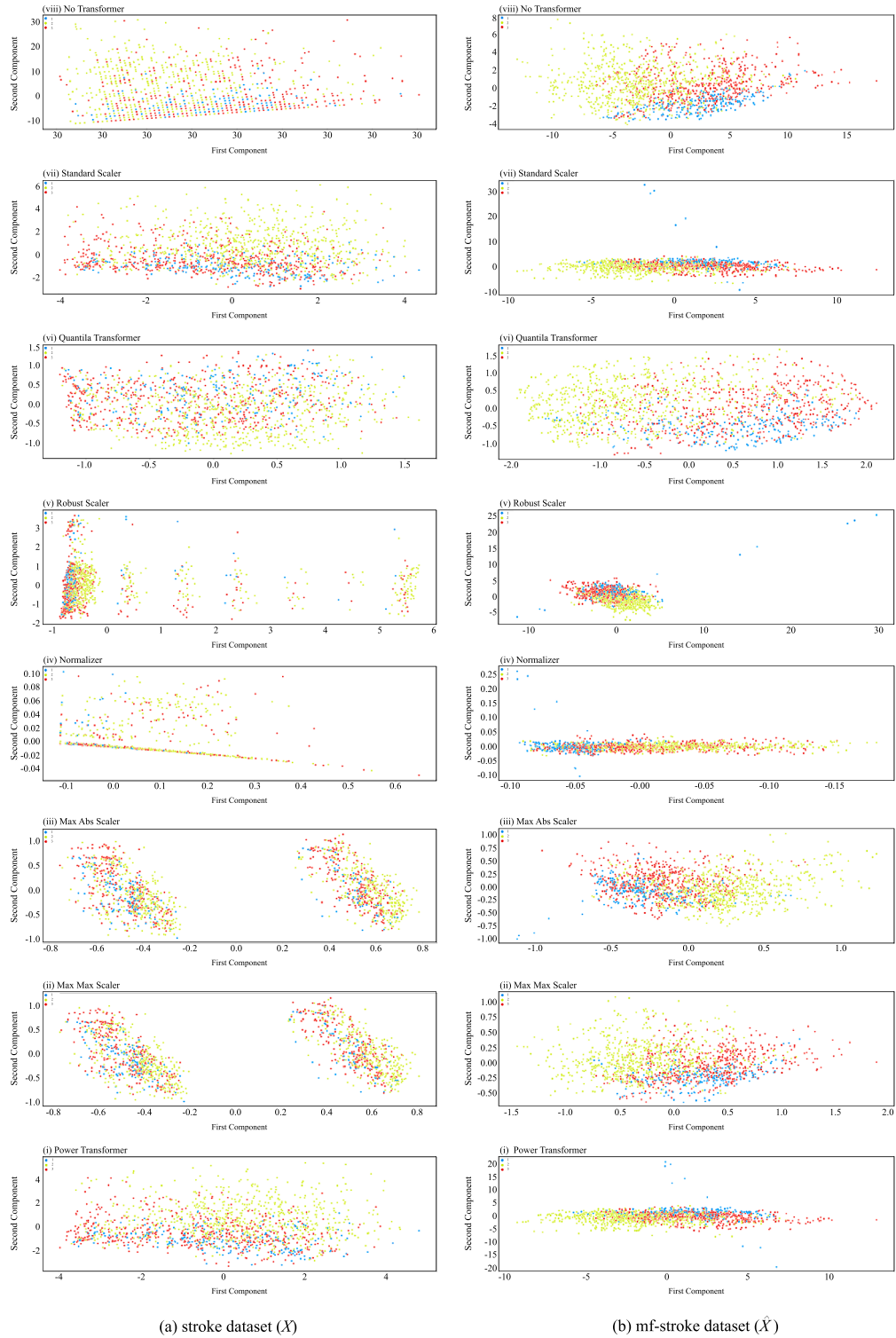


Fig. 6. Different scalers' model performance with PCA transformation for (a) stroke dataset (X) and (b) mf-stroke dataset ($\hat{X}$) for (i) Power Transformer (ii) Maximum Scaler (iii) Maximum Absolute Scaler (iv) Normalizer (v) Robust Scaler (vi) Quantile Transformer (vii) Standard Scaler (viii) No Transformer.

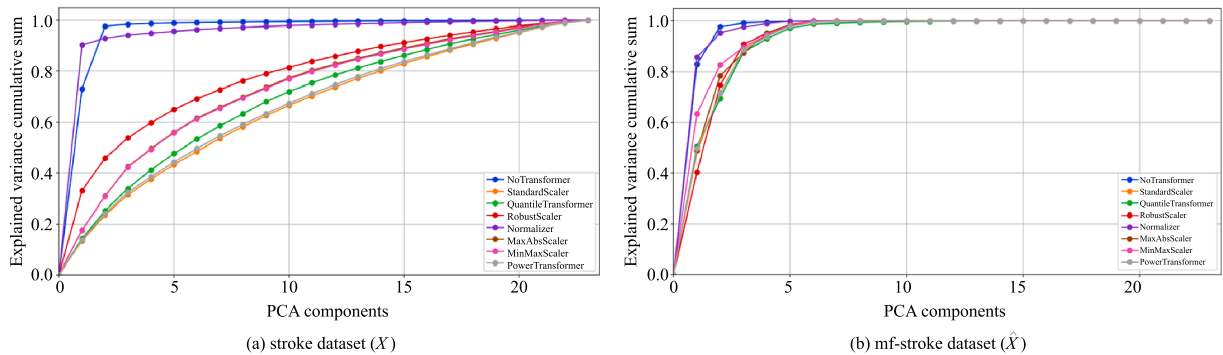


Fig. 7. Explained variance cumulative sum based on PCA components for (a) stroke dataset (X) and (b) mf-stroke dataset ($\hat{X}$).

For the classification and diagnosis of the three stroke subtypes, the related application steps of stacking generalization method (see Algorithm 3), which yielded the most robust and optimized prediction results on the stroke dataset (X) and mf-stroke dataset ($\hat{X}$), are provided in Algorithm 4.

Algorithm 4 The application steps of stacking generalization method on stroke dataset (X) and mf-stroke dataset ($\hat{X}$).

Input: Training Data $D = \{x_i, y_i\}_{i=1}^{1445}$ ($x_i \in \mathbf{R}^{1445}$, $y_i \in \{1, 2, 3\}$)

Output: An ensemble classifier H

Step 1: Divide the data set D into equally 5-fold subsets $\{D_1, D_2, \dots, D_K\}$

Step 2: for i from 1 to 5:

for j from 1 to 10:

Learn a base classifier h_{ij} from D/D_i

end

Step 3: After ending double for loop, we get the cross-validated predictions $p_1, p_2, \dots, p_{10}$, from all weak learners.

$p_j = \{h_{1j}(x_i), h_{2j}(x_i), \dots, h_{1445j}(x_i)\}$

Step 4: predictions $p_1, p_2, \dots, p_{10}$, from each of the l weak models' values will be combined to form a new (1445×10) matrix.

$$B = \left\{ \begin{bmatrix} p_1 \end{bmatrix}, \begin{bmatrix} p_2 \end{bmatrix}, \dots, \begin{bmatrix} p_{10} \end{bmatrix} \right\} \rightarrow \begin{bmatrix} p_1 & \dots & p_{10} \end{bmatrix}_{1445 \times 10}$$

Step 5: We call this new level-0 data.

Step 6: Train the meta learning algorithm h' (logistic regression) on a newly constructed level-0 data set.

Step 7: return $H(x) = h'(p_1, p_2, \dots, p_{10})$

The application of bagging, boosting and stacking methods, as the ensemble methods addressed in our proposed model, to both stroke dataset (X) and mf-stroke dataset ($\hat{X}$) is schematized in Fig. 7 (a) and Fig. 7(b), respectively. Bagging, boosting and stacking methods have yielded satisfactory results concerning the output classifications on mf-stroke dataset.

In the integrated multistage mathematical model proposed in our study, tree-based models enable the axis parallel boundaries sensitive to data rotation, providing an overfitting of the algorithm (see Fig. 7 (a) and (b)). By multifractal analysis, self-similar patterns are identified and significant attributes can be determined thereby. As one of the initial stages of the study, the PCA method has been employed to prevent the overfitting of the tree based classifiers, which, in turn, improved the robustness of the model through the multifractal analysis. The comparative aspect of this upper hand is depicted explicitly in Fig. 7 (a) and (b). Fig. 8 (a) and (b) presents the comparative analysis of the ensemble methods for (a) stroke dataset (X) (see Table 1 and 2 for the stroke dataset details) and (b) mf-stroke dataset ($\hat{X}$), respectively. The application of multifractal methods generated the mf-stroke dataset ($\hat{X}$) with irregular, self-similar, efficient and significant attributes the dataset having been identified by this initial method. The visual depiction of the results of the analyses is seen as distributions in the related figure with comparisons focusing on the application of the multifractal method.

Deep Learning methods (Feed Forward Neural Network, Multilayer Perceptron), ensemble methods (bagging methods (Random Forest and Extra Tree classifiers), boosting methods (XGBoost, Gradient Boosted, Light Gradient Boosting classifiers) and stacking method) and fundamental methods (Decision Tree, Logistic regression) and clustering methods (K-nearest neighbor, Support Vector Machine) have been applied to the stroke dataset (X) and mf-stroke dataset ($\hat{X}$) for the classification and diagnosis of three different stroke subtypes (small vessel, cardioembolic and cryptogenic) as a result of the application of the aspects in Table 7, which are based on the parameters in Table 5 and Table 6 in our integrated multistage mathematical model proposed.

The datasets formed are split as training data (924×24), validation data (232×24) and test datasets (289×24). The resampling technique is used to fix slight imbalances. The boosting algorithms generally yielded the highest results accordingly.

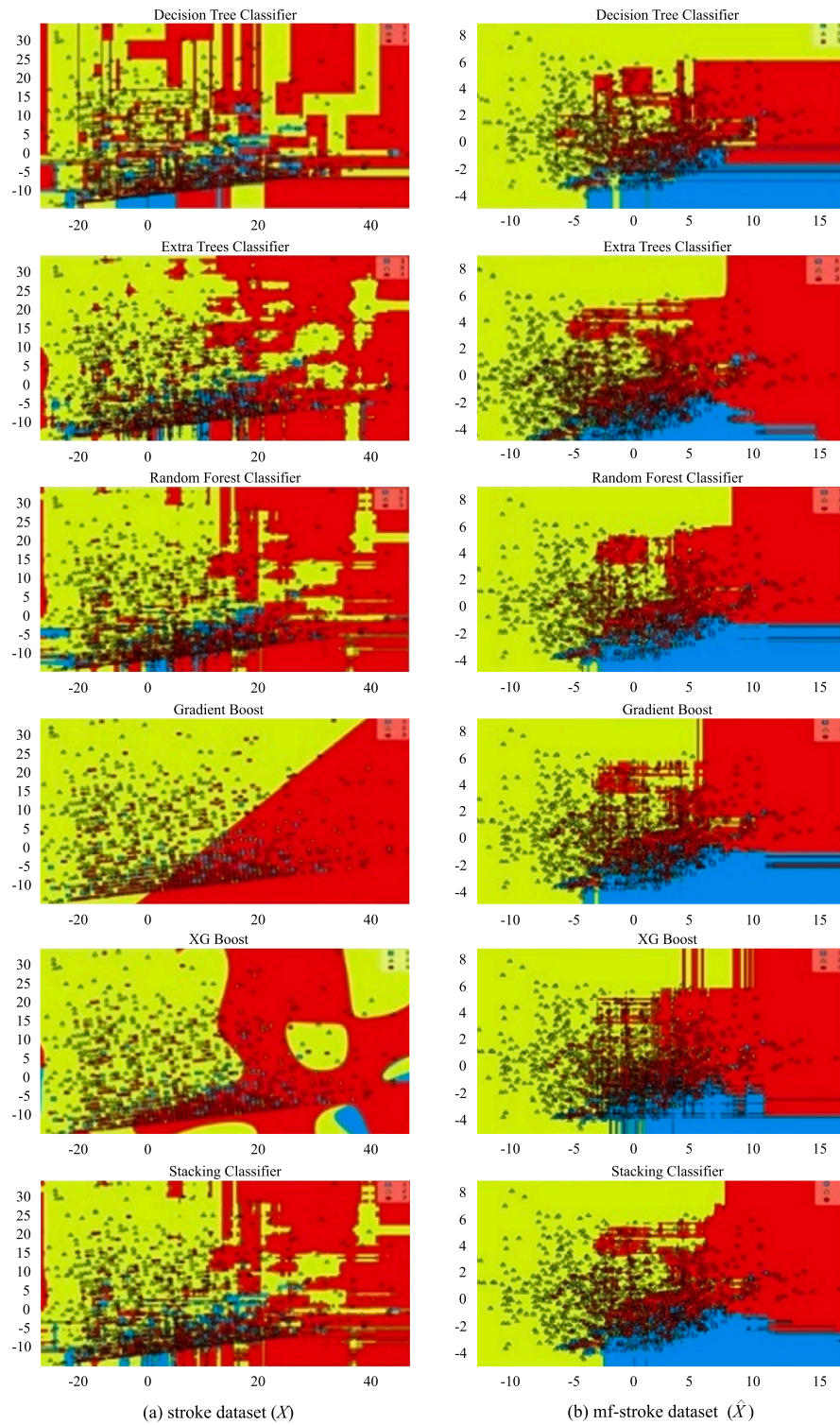


Fig. 8. Comparative analysis of the ensemble methods for (a) stroke dataset (X) and (b) mf-stroke dataset ($\hat{X}$).

Table 5

Network properties and values as obtained by bagging (RF), boosting (Xgboost) and deep learning (MLP).

Models properties/values	Random Forest (Bagging)	XGboost (Boosting)	MLP (Deep learning)
Activation Function	N/A	N/A	Relu
Solver	N/A	N/A	Adam
Performance Function	Categorical cross entropy	Categorical cross entropy	Categorical cross entropy
Learning Rate	N/A	0.1	Adaptive
Momentum	N/A	N/A	0.9
Class weights	Balanced	1	N/A
Number of hidden layers	N/A	N/A	6 with 128 neurons each
Number of estimators	100	100	N/A
Booster	N/A	Gbtree	Nesterovs Momentum
Objective	N/A	Multi: Soft Prob	N/A
Maximum depth	None	5	6
Criterion	Gini	N/A	N/A
Bootstrap	True	False	False

Table 6

Network properties and values of stacked generalization.

Model Properties	Values
Classifiers	Decision and Extra Trees, RF, KNN, SVM, MLP, XGBoost, LGBM, Logistic Regression, Gradient Boosting
Meta Classifier	Logistic Regression
Fit base estimators	True
Penalty	L2
Solver	lbfgs
Class weight	None
C value	1

Step 4: The comparative optimized prediction performances with accuracy rate results based on deep learning, ensemble methods, fundamental methods and clustering methods for the stroke subtypes' classification related to the stroke dataset (X) and mf-stroke dataset ($\hat{X}$)

In diagnostic complexity, multi-feature and parameters are at stake for the revealing of subtle and hidden patterns on a given dataset. To address such complexity and at the same time for the purpose of accurate treatment, timely intervention and saving lives ultimately, ten different algorithms, namely, deep learning methods (Feed Forward Neural Network, Multilayer Perceptron), ensemble methods (bagging methods (Random Forest and Extra Tree classifiers), boosting methods (XGBoost, Gradient Boosted, Light Gradient Boosting classifiers) and stacking method) and fundamental methods (Decision Tree, Logistic regression) and clustering methods (K-nearest neighbor, Support Vector Machine) have been applied to both the stroke dataset (X) and mf-stroke dataset ($\hat{X}$) in our study. By having two datasets, we developed a stacking generalization model using deep learning methods, ensemble methods, fundamental methods and clustering methods featuring scaled PCA to automatically predict stroke types more accurately based on medical history. As a result of all the comparisons of the methods, stacked generalization ensemble method with the application of multifractal regularization (L2 normalization) has yielded the highest accuracy results due to its being a high-level model to combine multiple models to achieve higher accuracy for predictive purposes in a classification task of all the three stroke subtypes displaying multifeatured and multivariate patterns.

Table 7 below shows the results of all implemented methods on test datasets of stroke dataset (X) and mf-stroke dataset ($\hat{X}$).

As Table 7 represents, Positive Predictive Value (PPV), Negative Predictive Value (NPV), F1 score (F1), Accuracy (ACC), Sensitivity (SN), Specificity (SP), Precision (PR), Recall (RC), Matthew's Correlation (MCC) were employed, and as a result, the PPV, NPV, F1, ACC, SN, SP, PR, RC, and MCC were calculated to classify the three stroke subtypes Small vessel, Cardioembolic, and Cryptogenic for both datasets, namely the stroke dataset (X) and mf-stroke dataset ($\hat{X}$)

As can be seen from the results for the classification of three subtypes of stroke in Table 7, the Stacking algorithm applied on the mf-stroke dataset ($\hat{X}$) outperforms compared to the stroke dataset (X) with all the ten related algorithms applied in our study. Stacking algorithm has achieved higher predictions through the reduction of the spread of the dispersion and reducing the variance component of the prediction error adding a bias.

Ensemble methods outperform other methods on both stroke and mf-stroke dataset. The bagging approach, one of the ensemble models, enables the decreasing of the classifier variance and reducing the overfitting. The other method, namely the boosting method, allows for the reduction of the bias and obtaining of the global minima on a complex gradient, generated by many categorical and numerical values. As one of the prominent AI techniques, ensemble learning, enables accumulation of decisions, uniting the predictions into an optimized definitive prediction process. In healthcare settings, identifying the patients' disease on time and taking the personalized, to-the-point and accurate measures will definitely help save their lives and maintain life quality on a sustainable scale. Consequently, this research's unified strategy and classification method can assist a facilitating plan to the physicians and

Table 7

Comparative optimized prediction results as obtained by deep learning methods, ensemble methods, fundamental methods and clustering methods for (a) stroke dataset (X) and for (b) mf-stroke dataset ($\hat{X}$).

(a) stroke dataset (X)												
Optimized Prediction Results	Deep Learning Methods		Ensemble Methods						Fundamental Methods		Clustering Methods	
	FFNN	MLP	Random Forest	Extra Trees	XGB	Grad. Boost	LGBM	Stacking	Dec. Tree	Logistic Reg	KNN	SVM
PPV	0.6302	0.63	0.7276	0.7337	0.7337	0.7235	0.7052	0.7378	0.693	0.5914	0.7276	0.7093
NPV	0.7851	0.815	0.8638	0.8668	0.8668	0.8617	0.8526	0.8689	0.8465	0.7957	0.8638	0.8546
F1	0.6302	0.63	0.7276	0.7337	0.7337	0.7235	0.7052	0.7378	0.693	0.5914	0.7276	0.7093
ACC	0.7201	0.7533	0.8184	0.8224	0.8224	0.8157	0.8035	0.8252	0.7953	0.7276	0.8184	0.8062
SN	0.6302	0.63	0.7276	0.7337	0.7337	0.7235	0.7052	0.7378	0.693	0.5914	0.7276	0.7093
SP	0.7851	0.815	0.8638	0.8668	0.8668	0.8617	0.8526	0.8689	0.8465	0.7957	0.8638	0.8546
PR	0.6302	0.63	0.7276	0.7337	0.7337	0.7235	0.7052	0.7378	0.693	0.5914	0.7276	0.7093
RC	0.6302	0.63	0.7276	0.7337	0.7337	0.7235	0.7052	0.7378	0.693	0.5914	0.7276	0.7093
MCC	0.4506	0.4656	0.5772	0.5751	0.5837	0.5639	0.5422	0.5928	0.5075	0.3781	0.5793	0.5588
(b) mf-stroke dataset ($\hat{X}$)												
Optimized Prediction Results	Deep Learning Methods		Ensemble Methods						Fundamental Methods		Clustering Methods	
	FFNN	MLP	Random Forest	Extra Trees	XGB	Grad. Boost	LGBM	Stacking	Dec. Tree	Logistic Reg	KNN	SVM
PPV	0.7508	0.8824	0.8894	0.8824	0.894	0.894	0.8847	0.9032	0.7926	0.7396	0.7534	0.7235
NPV	0.8754	0.9412	0.9447	0.9412	0.947	0.947	0.8423	0.9516	0.8963	0.8698	0.8767	0.8617
F1	0.7508	0.8824	0.8894	0.8824	0.894	0.894	0.8847	0.9032	0.7926	0.7396	0.7534	0.7235
ACC	0.8339	0.9216	0.9262	0.9216	0.9293	0.9293	0.9231	0.9354	0.8617	0.8264	0.8356	0.8156
SN	0.7508	0.8824	0.8894	0.8824	0.894	0.894	0.8847	0.9032	0.7926	0.7396	0.7534	0.7235
SP	0.8754	0.9412	0.9447	0.9412	0.947	0.947	0.9423	0.9516	0.8963	0.8698	0.8767	0.8617
PR	0.7508	0.8824	0.8894	0.8824	0.894	0.894	0.8847	0.9032	0.7926	0.7396	0.7534	0.7235
RC	0.7508	0.8824	0.8894	0.8824	0.894	0.894	0.8847	0.9032	0.7926	0.7396	0.7534	0.7235
MCC	0.6099	0.2834	0.8206	0.8103	0.829	0.8287	0.8145	0.844	0.6693	0.5963	0.601	0.5912

clinicians concerning any disease or medical incident, especially relevant to the brain, which is characterized by a highly complex heterogeneous and complicated dynamics as well as attributes.

4. Conclusion and future outlooks

Mathematical modeling based on modeling neuron and neural network dynamics in conjunction with artificial intelligence (AI) methods can be employed for capturing the processes of the functioning of neural networks along with their subtle features in complex systems that are often maintained by flows of information. High degrees of complexity in various areas, including different debilitating and life-threatening medical conditions and concerns particularly, have compelled the emergence of applications to be more intelligent for the related critical decision-making processes. Consequently, complex systems research and AI display a close interconnection, which presents a common denominator for a comprehensive range of interdisciplinary applications and interactions. The integrated multistage mathematical model proposed in this study aims to allow for the intersection of deep learning, machine learning and ensemble methods in AI together with the singularity analysis in order to extract subtle patterns in the stroke dataset by multifractal regularization (L2 normalization). The originality of this study is due to the gap in the literature where a multistage combination approach with algorithms, made up of deep learning methods (Feed Forward Neural Network, Multilayer Perceptron), ensemble methods (bagging methods (Random Forest and Extra Tree classifiers), boosting methods (XGBoost, Gradient Boosted, Light Gradient Boosting classifiers) and stacking method) and fundamental methods (Decision Tree, Logistic regression) and clustering methods (K-nearest neighbor, Support Vector Machine) as well as multifractal regularization (L2 normalization) have been employed on a multifeatured complex stroke dataset (X). Regarding the approach-wise novel points of in this study, the following can be noted: First of all, multifractal regularization (L2 normalization) method has been applied to the stroke dataset (X) (see Table 2) and thereby, the mf-stroke dataset ($\hat{X}$) was generated, through which the distance was minimized, and calculations were simplified as per the dataset with a high degree of complexity. In this way, irregular, self- similar, efficient and significant attributes in stroke dataset have been identified by this initial method. Secondly, the algorithms consisting of the deep learning methods, ensemble methods, fundamental methods and clustering methods have been applied both to the stroke dataset (X) and the mf-stroke dataset ($\hat{X}$). Thus, assisting the accurate individualized treatment and intervention for stroke cases characterized by complexity and unpredictability has been enabled through this stage, which is highly critical for the timely and accurate determination of the stroke subtypes. When the current study is compared with the other previous related ones [49,59,73–77,79,81] conducted thus far, the present study has provided these related novelty points. Eventually, as the last stage, the accuracy rates have been compared along with the specificity,

sensitivity, precision, F1-score and Matthews Correlation Coefficient analyses, following the application of the ensemble methods and deep learning methods (see Table 7), which are indicated in the second stage. The results with the integrated multistage proposed mathematical model have demonstrated that stacking method, one of the related ensemble methods, has yielded results with a higher level of accuracy as the singularity analysis has been performed on the mf-stroke dataset ($\hat{X}$), through which it has been possible to detect the significant and self-similar attributes of the stroke dataset (X). The experimental results obtained demonstrate the applicability, accuracy and validity of the proposed model, while emphasizing the critical aspect of the significant attributes for the diagnosis and determination of the diseases, which is highly important for timely and correct decision-making process, considering the unpredictable, irregular, self-similar and transient attributes of stroke besides other neurological diseases.

Relying on these related respects, the following directions and outlooks may be presented for future research:

- (1) By the demonstration that stacking generalized method, among the related ensemble methods, has yielded more accurate results as the singularity analysis has been performed on mf-stroke dataset ($\hat{X}$), it has been revealed that significant and self-similar attributes have been able to be detected. This also reveals the advantage of ensemble methods which are capable of learning more complex decision and hypothesis with a higher level of accuracy. This aspect and advantage can indicate a way for future studies which may focus on accuracy in critical decision-making processes.
- (2) The study also shows the importance of utilization of deep learning methods which employ a specific type of AI architecture, named artificial neural networks (ANN), which emulate the humans' brain functions. This direction is important for related studies concerning other neurological and brain-related diseases in medicine, neuroscience, biology, neuromorphic computing, to name some.
- (3) The integrated multistage mathematical model proposed in this study can provide a structured direction for researchers to focus on the way of selecting the significant attributes and reducing overfitting in the stroke dataset for diagnostic processes, detection, identification and classification purposes based on the grounds of advanced computing system applications where human knowledge, intuition, experience and heuristics are essential.
- (4) The proposed method provided in this study may be taken as a model to set up interface(s) for models used like the ones employed in [116,117] in the future.
- (5) The experimental results obtained by this study and the integrated multistage mathematical model proposed by this study can provide an avenue for the detection, classification, diagnosis as well as optimized outcome prediction of diseases which manifest a high level of complexity and heterogeneity.
- (6) The integrated multistage mathematical model of the current study can set an example for future research which can address the minimization of the cost burden of brain stroke detection as well as other neurological diseases by taking into consideration the data-driven approaches.
- (7) The scheme proposed by this study can provide a novel way for future studies and projects which aim at merging artificial intelligence methods and advanced mathematical models in complex settings.

Taken together, the present study has aimed to underpin a novel direction in the related domains regarding dynamic and complex systems. The experimental results obtained validate the applicability and validity of the proposed method in which deep learning methods, ensemble methods, fundamental methods and clustering methods of AI along with multifractal analysis facilitate and empower the development and application of alternative applicable models in multifaceted healthcare systems. Furthermore, the use of multistage mathematical model in our study shows that accuracy is not the only parameter to evaluate the significance of a particular technique. Rather, multi-criteria decision making should also be considered, which constitutes the essence of our proposed complementary and integrative multistage framework. Through a different and novel outlook of AI and computing system applications, it has been possible to tackle the establishment and maintenance of personalized treatment and timely management of the stroke and its subtypes which manifest characteristics of uncertainty, unpredictability, variability, dynamism and complexity.

Data availability

The data that has been used is confidential.

Declaration of competing interest

The authors declare that they have no conflict of interest.

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